Female Urology

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · **P2** = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 101	Surgical, Radiographic and Endoscopic Anatomy of the Female Pelvis
Ch 108	Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology
Ch 105	Evaluation and Management of Women With Urinary Incontinence and Pelvic Organ Prolapse
Ch 114	Conservative Management of Urinary Incontinence: Behavioural and Pelvic Floor Therapy, Urethral and Pelvic Devices
Ch 116	Retropubic Suspension Surgery for Incontinence in Females
Ch 118	Slings: Autologous, Biologic, Synthetic and Mid- Urethral
Ch 117	Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse
Ch 119	Genitourinary Complications of Mesh and Their Repair
Ch 121	Urinary Tract Fistulae
Ch 122	Bladder and Female Urethral Diverticula
Ch 77	Sexual Function and Dysfunction in the Female
★	Putting it together: female pelvic decisions

101 Surgical, Radiographic and Endoscopic Anatomy of the Female Pelvis

1 · PERINEUM AND EXTERNAL GENITALIA

- The meatus lies **1 cm** anterior to the vaginal opening.
- **Skene's (paraurethral) glands** sit beside the meatus and are the prostate homologue; obstructed periurethral glands form a diverticulum or cyst.
- **Bartholin's gland** lies under the posterior end of each vestibular bulb and may form a cyst or abscess.
- The **deep perineal space** holds the urethral sphincter, urethrovaginal sphincter, compressor urethrae and deep transverse perineal; the **superficial space** (under Colles fascia) holds the superficial perineal muscles, clitoris, vestibular bulbs and Bartholin's glands.
- The perineal muscles and external anal sphincter converge at the **perineal body**, so episiotomy is angled laterally to spare the sphincter.
- The **labia majora have a dual supply** (superficial external pudendal from above, posterior labial from below), so the **Martius flap** is reliable.

2 · BONY PELVIS AND LIGAMENTS

- The **ischial spine** marks apical sutures.

Fixation point	Used for	Adjacent danger
Sacrospinous ligament (SSL): ischial spine to lower sacrum/coccyx	SSL fixation (apical prolapse)	Pudendal nerve/vessels near the spine; sacral plexus just posterior

Fixation point	Used for	Adjacent danger
Cooper's ligament (medial superior pubic ramus)	Burch colposuspension for stress urinary incontinence (SUI)	Obturator vessels; aberrant obturator
Obturator foramen; canal superolateral	Transobturator tape (TOT) trocar	Obturator nerve/vessels, causing groin pain or bleeding
Anterior longitudinal ligament (ALL) over S1	Sacrocolpopexy mesh	Presacral veins, middle sacral vessels, left common iliac vein, right ureter
Uterosacral ligament	Uterosacral (apical) suspension	Ureter (lateral, closest at the cervix); sacral nerve entrapment

- The pudendal nerve leaves through the **greater** sciatic foramen and re-enters through the **lesser** to reach Alcock's canal.

3 · PELVIC FLOOR MUSCLES

- The pelvic diaphragm is the **levator ani** and coccygeus; the levator is supplied directly by **S3–S5 (mainly S4)**.

P1 *Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — levator table: puborectalis, pubococcygeus, iliococcygeus; coccygeus NOT levator, lies on SSL*

Not in the paper answer:

- The **levator hiatus** transmits the urethra, vagina and rectum, and the **levator plate** is a horizontal visceral shelf; injury leaves the hiatus lax, the plate tilts and **prolapse** follows.

4 · ENDOPELVIC FASCIA AND VISCERAL LIGAMENTS

- The **intermediate** fascial stratum (parametrium, paracolpium, pubocervical fascia) **gives most organ support**.
- The **pubourethral ligaments** (puboprostatic homologue) and anterior vaginal wall form a mid-urethral hammock; weakness causes SUI.
- A vesicopelvic fascia defect at the arcus tendineus gives a paravaginal cystocele.
- The **cardinal (Mackenrodt's) ligament** carries the pelvic plexus, so dividing it at hysterectomy can cause bladder dysfunction.
- Distinguish the **infundibulopelvic ligament** (ovarian vessels, ureter beneath) from the **ovarian ligament** (ovary to uterus) MCQ.

DeLancey levels of vaginal support †

P1 *Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — I suspension (uterosacral–cardinal; uterine/vault prolapse, enterocele), II attachment (ATFP; anterior/posterior wall), III fusion (urethral hypermobility, distal rectocele, perineal descent)*

> **KEY POINT**
>
> Support ligaments are connective tissue condensed around vessels and nerves, so the best anchors (uterosacral, cardinal) are exactly where sutures injure the ureter, pelvic plexus and sacral nerves. See Ch 108

5 · URETHRA AND CONTINENCE ANATOMY

- The urethra is **4.7–5.1 cm** long and hangs from the posterior pubourethral ligaments on the anterior vaginal wall.
- **The mucosa and spongy vascular submucosa generate most closure pressure and are oestrogen-dependent**, which is the rationale for vaginal oestrogen.
- The **internal sphincter** is smooth and autonomic at the bladder neck; the **external sphincter** (sphincter urethrae, **compressor urethrae**, **urethrovaginal sphincter**) is striated, voluntary

(pudendal) and covers the distal two-thirds.

- **Hypermobility** is one cause of SUI. An autologous pubovaginal sling sits at the **bladder neck**; a synthetic mid-urethral sling (MUS) sits at the **mid-urethra**.

6 · BLADDER AND URETER

The ureter is the structure most often injured.

- The **dome** is peritonealised, so intraperitoneal injury needs formal repair. Injury to the **base** causes vesicovaginal or vesicouterine fistula.
- The **pelvic ureter** crosses the iliac bifurcation, passes behind the ovary, runs **under the uterine artery** ("water under the bridge"), then crosses the anterior fornix to the bladder.

ALGORITHM

1. **Ureteric danger points**:
2. Pelvic brim, at the infundibulopelvic ligament.
3. Under the uterine artery (hysterectomy).
4. Beside the cardinal/uterosacral ligaments near the cervix.
5. Anterior vaginal fornix (vaginal surgery).
6. Sacral promontory: the right ureter (sacrocolpopexy).

7 · NERVES AT SURGICAL RISK

- The **pudendal nerve (S2–S4)** crosses the SSL **close to the ischial spine** into Alcock's canal; pudendal block is given transvaginally just behind the spine.
- In SSL fixation, place sutures about 2 cm medial to the spine, through (not around) the ligament †.

Nerve	Procedure / mechanism	Deficit
Common peroneal	Lithotomy: compression at the fibular head	Foot drop
Sacral plexus (L4–S4)	SSL fixation (plexus exits just posterior to the SSL)	Posterior leg motor/sensory loss
Pudendal	SSL fixation near the spine	Perineal pain, sensory loss, sphincter dysfunction
Sacral trunks S1–S4	Uterosacral suspension (deep sutures)	Buttock/posterior thigh pain
Ilioinguinal (L1), genitofemoral (L1–L2)	Hernia repair, caesarean, hysterectomy	Mons/labial numbness or pain
Obturator (L2–L4) †	TOT trocar, obturator lymphadenectomy	Groin pain, adductor weakness
Posterior femoral cutaneous (S1–S3)	Gluteal pressure, pelvic tumour	Perineal sensory loss

KEY POINT

Post-lithotomy foot drop is peripheral (fibular head), not pelvic. Leg pain and swelling after long lithotomy suggest **compartment syndrome**, an emergency († fasciotomy).

8 · VESSELS AND LYMPHATICS

Internal iliac	Anterior division	Posterior division
Visceral	Obliterated umbilical, superior vesical, uterine, vaginal, middle rectal	None
Parietal	Obturator, internal pudendal, inferior gluteal	Iliolumbar, lateral sacral, superior gluteal

- The **internal iliac** has rich collaterals, so ligation or embolisation is tolerated; ligating the **external iliac** threatens the limb.
- The **presacral plexus** is valveless and under high pressure, and a 1–2 mm vein can bleed more than **1 L/min**.

ALGORITHM

1. **Sacrocolpopexy**
2. Expose the ALL at the promontory over the L5/S1 disc.
3. Open the peritoneum distally.
4. Suture into the ALL over **S1** (strongest; less discitis).
5. Avoid the left common iliac vein, middle sacral vessels, presacral plexus and right ureter (each about 2.7 cm from the mid-promontory).

Organ	First nodes
Distal ⅓ urethra, lower ⅓ vagina, vulva	Inguinal, then pelvic
Proximal ⅔ urethra, upper ⅔ vagina	Iliac, obturator (pelvic)
Bladder, uterus	Perivisceral, then pelvic, presacral and great-vessel nodes
Ovary MCQ	Along the gonadal vessels to para-aortic nodes (bypasses the pelvis)

9 · ENDOSCOPIC ANATOMY

ALGORITHM

1. Distal urethra: glands, diverticular ostia, mesh, stones.
2. Mid-urethral coaptation (external sphincter).
3. Bladder neck coaptation.
4. Trigone and orifices (30° lens).
5. Survey the whole mucosa.

- Squamous metaplasia and cystitis cystica are benign, but biopsy if in doubt.

10 · IMAGING

- **Translabial ultrasound** shows bladder-neck mobility, prolapse, and **mesh, MUS and bulking agents**.
- On **endoanal ultrasound**, an internal sphincter defect is a bright break in dark muscle; an external sphincter defect is a dark lesion in bright muscle.
- **T2-weighted MRI is best for urethral diverticulum** and separates other cysts such as Gartner, Skene and Bartholin cysts.
- **MR defecography** grades descent below the **pubococcygeal line** and is used for defaecatory dysfunction or symptoms out of proportion to examination.

108 Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology

1 · DEFINITIONS AND TERMINOLOGY

The type of UI decides treatment.

- **Urinary incontinence (UI)** is complaint of involuntary urine loss (ICS/IUGA, 2010).
- Symptoms are non-specific, so combine them with signs and observations (diary, urodynamics [UDS]). **Bother drives treatment**: without quality-of-life impact, none may be needed.

Type	Complaint of loss…	Urodynamic correlate / trap
Stress UI (SUI)	On effort, cough, sneeze	Urodynamic SUI: leak with abdominal pressure rise **and no detrusor contraction**
Urgency UI (UUI)	With urgency	**Detrusor overactivity (DO) not required**; DO incontinence (DOI) is a separate UDS diagnosis
Mixed UI (MUI)	With urgency and effort	Urge- or stress-predominant, or equal
Overflow UI	Frequent small-volume loss with retention	Underactivity or obstruction; high post-void residual (PVR)
Functional UI †	Impaired mobility/cognition, normal lower urinary tract	Frail elderly, long-term care
Continuous UI	Day and night, few voids	Suspect fistula
Nocturnal enuresis	During sleep	Differs from nocturnal UUI (wakes, cannot reach toilet)

- **Urgency** is a sudden compelling desire that is hard to defer (pathological), whereas **urge** is normal desire.
- **Overactive bladder (OAB)** is urgency with or without UUI, usually with frequency and nocturia; it is **defined by urgency**. Frequency ≥ **8 voids/24 h** is abnormal once polyuria is excluded.

2 · EPIDEMIOLOGY OF FEMALE UI

- Remission reaches about 12%/year and is **lowest for MUI**.
- **Pregnancy** is the commonest cause of new UI in young women (about **40%**, mostly SUI). UI at 3 months postpartum usually persists, and pelvic floor muscle training (PFMT) during and after pregnancy reduces it.
- **Hysterectomy** can cause or cure UI; vaginal and laparoscopic routes carry more risk than open.

3 · RISK FACTORS FOR FEMALE UI

Factor	Key point
Ageing	Strongest independent non-modifiable factor. SUI peaks at 40–59 years; UUI/MUI dominate after 60. Over 40% in long-term care at admission
Vaginal delivery	Causes later **SUI, not UUI**. Caesarean section (CS) protects only if **every** delivery is by CS; number needed to treat (NNT) is **13** to prevent one moderate–severe UI. Elective CS protects more than CS in labour. Baby ≥4 kg: OR about 1.5
Parity	First delivery OR about 1.5, then modest rise; strongest for SUI
Obesity	Body mass index (BMI) ≥30 more than doubles risk (SUI, MUI); **reversible** with weight loss. Little effect on SUI surgery success
Menopause / oestrogen	**Systemic oestrogen increases UI** (new UI nearly doubled vs placebo). Vaginal oestrogen is not linked to SUI and treats atrophy and recurrent UTI
Race	SUI commoner in White than Black/Asian women (higher closure pressure in Black women)
Smoking	Urgency, frequency, UUI (dose-dependent)
Caffeine	UUI, MUI, OAB, but not SUI
Diabetes	Up to 2×; neuropathy and macroalbuminuria are independent factors

> **KEY POINT**
>
> Obstetric injury and loss of support cause **stress** UI; ageing, neurological, metabolic and lifestyle factors cause **urgency**. So the dominant type moves from stress to urge with age.

Male UI (contrast)

- In men **UUI is commoner than MUI, which is commoner than SUI**. Male SUI is nearly always iatrogenic (radical prostatectomy [RP]) or neurological, and remission reaches **40%**.
- Post-RP UI is 8–60% depending on definition; after TURP it is about 1%.

4 · PHYSIOLOGY OF CONTINENCE

- During filling, **pelvic** parasympathetic outflow is suppressed, **hypogastric** sympathetic β-action relaxes the detrusor and α-action closes the bladder neck, and the **pudendal** nerve (Onuf's nucleus) keeps the sphincter closed.
- In **women**, the continence zone is the **proximal and mid-urethra**; the bladder neck is often incompetent in continent women. The oestrogen-sensitive spongy submucosa gives up to **30%** of closure.
- **Support**: the anterior vaginal wall is the firm backing. Lax vaginal support with intact pubourethral ligaments causes **shear** in the continence zone.
- **Male vs female**: men tolerate losing one outlet unit (TURP) because the bladder neck is powerful; in women the bladder neck is weak, so external sphincter deficiency causes UI.

> **KEY POINT**
>
> The female outlet relies on a **single** mid-urethral sphincter against a mobile backing, so continence needs support, intrinsic urethral properties and muscle together.

5 · GENERAL PATHOPHYSIOLOGY: BLADDER VS OUTLET

Mechanism	Causes	Result
Loss of suprapontine inhibition	Stroke, multiple sclerosis (MS), Parkinson; early diabetes mellitus (DM)	Neurogenic DO, causing UUI
Outlet obstruction	Women: after anti-incontinence surgery; men: bladder outlet obstruction (BOO)	DO with new (de novo) UUI
Detrusor underactivity	Conus/cauda lesion, neuropathy (DM, alcohol), radical hysterectomy, abdominoperineal resection (APR), opioids including tramadol	Retention, then **overflow UI**
Poor compliance	Pelvic radiotherapy	High detrusor leak point pressure (LPP), causing UI
Sphincter deficiency	RP, urethral/anti-incontinence surgery, cord injury, α-blockers, muscle relaxants	Intrinsic sphincter deficiency (ISD), causing SUI

- In **DM**, early polyuria and UUI give way to lost sensation and contractility, with poor emptying, UTI and overflow.
- In multiple system atrophy ≥20% leak after TURP.
- In post-RP UI, **ISD predominates (>70%)**, and DO or poor compliance contributes in >30%.
- Each failed operation adds fibrosis and ISD; **labour** causes fascial injury and **pudendal denervation**.

6 · PATHOPHYSIOLOGY OF FEMALE SUI

P3 P1 *Female pelvic floor, continence mechanism, classification and management of SUI — theories table: pressure transmission (Enhorning), hammock (DeLancey; repositioning not essential), integral theory (Petros–Ulmsten, giving MUS), ISD (McGuire)*

Not in the paper answer:

- Support does not explain everything: some women with marked hypermobility are dry, and some with a fixed urethra leak severely.

Hypermobility vs ISD

- ISD equals **type III SUI** (Blaivas) MCQ and is usually seen after previous surgery.
- A **"pipestem" urethra** is fixed and non-coapting; subtler ISD coexists with hypermobility in most SUI.
- **Valsalva leak point pressure (VLPP)** is the abdominal pressure that causes leakage. **VLPP < 60 cm H_2O means ISD**, historically treated by bulking or pubovaginal sling.

> **KEY POINT**
>
> **Slings work in hypermobility but Burch fails in ISD**. Almost all SUI includes some ISD, so sphincter-directed procedures (pubovaginal sling, mid-urethral sling [MUS], bulking) work in hypermobile women. Burch corrects only hypermobility, so it is weak with severe ISD and a fixed urethra.

7 · OAB PATHOPHYSIOLOGY (BRIEF)

- Neurogenic causes (§5) and outlet obstruction (BOO, post-sling) both lead to DO and de novo urgency.
- With pelvic organ prolapse (POP), urgency is mainly due to obstruction from kinking. **OAB improves after POP repair in 60–80%**, more after anterior/apical repair, with no reliable predictor.

8 · INSENSIBLE INCONTINENCE

- It is usually ISD or DOI; if treatment fails, seek rarer causes. Post-void dribble suggests diverticulum (pelvic MRI), continuous loss with normal voids suggests ectopic ureter (upper-tract imaging), and frequent small losses suggest overflow (UDS and PVR).

9 · PELVIC ORGAN PROLAPSE: DEFINITION AND STAGING

- **POP** is descent of the uterus/vault and/or vaginal walls from loss of support. **Name the vaginal segment** (anterior, apical, posterior), not the presumed organ.
- The most specific symptom is a **visible or palpable bulge**, which correlates best once past the hymen.
- Never attribute bleeding or pain to POP; relief with a **pessary trial** predicts benefit from surgery.

POP-Q stage	Most distal point on maximal strain
0	Aa, Ba, Ap, Bp = −3; C/D ≤ −(tvl − 2)
1	> 1 cm above hymen (< −1)
2	Within 1 cm of hymen (−1 to +1)
3	> +1 but < +(tvl − 2)
4	Complete eversion: ≥ +(tvl − 2)

P1 *Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — POP-Q points: hymen 0; Aa/Ap 3 cm (−3 to +3), Ba/Bp, C, D, gh, pb, tvl; 3 × 3 grid*

Not in the paper answer:

- The stage is set by the single most distal point, and measuring right after pessary removal underestimates it.
- POP-Q is the IUGA/ICS-recommended system; Baden–Walker grades have poor interobserver reliability.

10 · POP EPIDEMIOLOGY AND RISK FACTORS

- Loss of support is seen in 43–90%, and **asymptomatic stage 1–2 is normal**; symptomatic POP affects 3–12%.
- **Anterior POP is commoner than posterior, which is commoner than apical**. Apical descent accompanies most significant anterior POP and is universal at stage 4. **So when planning cystocele repair, assess the apex and consider vault suspension.**
- POP progresses slowly; progression predictors are age >60, Ba ≥4 cm and genital hiatus (gh) ≥5 cm.

Risk factor	POP	UI (contrast)
Vaginal childbirth	Strongest; rises steeply with parity; forceps worse. Elective CS risk is about nulliparous; CS in labour about vaginal	SUI, not UUI
Age	About 40% rise per decade; surgery peaks in the 7th decade	Shift to UUI/MUI
Obesity	About 1.4–2.5-fold	>2-fold, reversible
Hysterectomy	Raises risk, most if done for POP (no vault suspension)	Cause or cure
Race	Hispanic highest, Black lowest	White highest
Oestrogen / menopause	Not consistently associated	Oral oestrogen increases UI
Other	Family history (about 5× in siblings), connective-tissue disease, chronic constipation (2.5×)	—

- Recurrence after surgery is predicted by preoperative stage 3–4.

11 · PATHOPHYSIOLOGY OF POP

- The **levator ani** is the primary support; the uterosacral–cardinal ligaments hold the organs in position over it.

P1 *Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — DeLancey I suspension (uterine/vault, enterocele), II attachment to ATFP (lateral or central anterior defect), III fusion (urethral hypermobility, distal rectocele)*

- The commonest cystocele defect at surgery is detachment of the arcus tendineus fasciae pelvis (ATFP) from the ischial spine.

ALGORITHM

1. **POP sequence**
2. Labour causes levator trauma or denervation.
3. The muscle loses tone and the urogenital hiatus widens.
4. Fascia and ligaments now carry the whole load.
5. They stretch and fail (ageing, straining, hysterectomy): POP.
6. POP widens the hiatus further.

- MRI shows levator defects in up to **20%** of primiparas and none in nulliparas; **avulsion** predicts advanced, multicompartment POP.
- Levator neuropathy follows vaginal delivery or CS in labour but almost never elective CS, so **labour itself is the injury**.
- An altered collagen type I:III ratio and Ehlers–Danlos or Marfan syndrome increase risk.

12 · POP AND UI INTERACTION

Prolapse can hide SUI until repair.

- 40–55% of women with SUI have significant anterior POP, as do about 20% with OAB.
- **Occult SUI** is SUI seen only with the prolapse reduced (up to a quarter of advanced POP).

ALGORITHM

1. A cystocele kinks the urethra, and the obstruction masks SUI.
2. POP repair relieves the kink, so SUI is unmasked.
3. Detect it by UDS or stress test **with the prolapse reduced** (pessary or speculum).

- De novo SUI after POP surgery is predicted by age >66, DM and low urethral closure pressure.
- Examine for POP in every UI evaluation, because POP missed at sling surgery means later prolapse surgery.

13 · LANDMARK TRIALS

Trial	Design	Result	Take-home
CARE	Stress-continent women, abdominal sacrocolpopexy with or without Burch	Burch reduced postoperative SUI; SUI on UDS with prolapse reduced predicted it	Prophylactic Burch at sacrocolpopexy prevents de novo SUI
OPUS	Stress-continent women, vaginal POP repair with prophylactic MUS vs sham	UI at 1 year 27% vs 43% (NNT about 6); more bladder perforation, UTI, bleeding, retention	Fewer de novo SUI, more adverse events, so counsel

105 Evaluation and Management of Women With Urinary Incontinence and Pelvic Organ Prolapse

1 · DEFINITIONS AND RISK FACTORS

Anatomical prolapse is common; disease means symptoms.

- Risk factors for **urinary incontinence (UI)** are age, obesity, smoking, comorbidity and postmenopausal hormones; vaginal delivery causes stress UI (SUI).
- **Pelvic organ prolapse (POP)** is commonest anteriorly, then posteriorly, then apically. Half of parous women have anatomical POP, but disease means **symptoms attributable to it**. A **vaginal bulge** is the most specific symptom.

Category	POP risk factors
Primary only	Older age, high body mass index (BMI), high birth weight, levator defects, low education, poverty
Recurrence only	**Younger age**, preoperative Pelvic Organ Prolapse Quantification (POP-Q) stage 3–4
Both	Vaginal delivery, parity, levator hiatal ballooning
Protective (primary)	Caesarean delivery; **smoking** MCQ

- **ICS UI terms** follow the trigger: SUI, urgency UI (UUI) and mixed UI (MUI). **Insensible** UI occurs without awareness, and **occult SUI** appears only after POP reduction.

2 · HISTORY

- Ask directly and exclude reversible causes with **DIAPPERS**: Delirium, Infection, Atrophic vaginitis, Psychological, Pharmacological, Excess urine output, Restricted mobility, Stool impaction.
- **SUI is urethral failure, UUI is bladder failure, and MUI is both**. In MUI, treat the more **bothersome** component first.
- **POP** gives a bulge and pressure that worsen later in the day. **Splinting** suggests high-grade POP. Anterior POP kinks the urethra and causes voiding difficulty; posterior POP obstructs defecation.
- SUI coexists in about **40%** of women with POP, and many expect repair to cure leakage, so counsel them.

- **Occult SUI** is found in about ⅓ before vaginal POP repair, and **de novo SUI** can appear after POP surgery; both need a shared decision on concurrent versus staged surgery.
- Ask about neurological disease, diabetes, radiation, pelvic or mesh surgery and obstetric trauma. **Local** oestrogen may improve UI and **systemic** oestrogen may worsen it.
- **Drugs**: **α-blockers worsen SUI in women**, and α-agonists, anticholinergics and opioids impair emptying. Diuretics, SGLT2 inhibitors and lithium cause polyuria. **ACE inhibitors cause cough, which worsens SUI**.

3 · PHYSICAL EXAMINATION

- Atrophic (oestrogen-deficient) tissue is pale, flat, dry and without rugae.

SUI testing

- AUA/SUFU 2023 says: **objectively demonstrate SUI with a comfortably full bladder before surgery**.

P3 *SUI: evaluation in females, role of UDS, and surgical therapy in males — cough stress test (supine, then standing), reduce prolapse for occult SUI, Q-tip >30° = hypermobility*

Not in the paper answer:

- The Q-tip test is now mostly for teaching and trials; **visual urethral mobility** (meatal rotation < or **>30°**) is routine.

POP examination

- Examine in lithotomy **and standing**, while straining, to see the worst prolapse.
- Use a **split Graves speculum** to isolate each compartment in turn. If the blade is expelled with posterior foreshortening, there is **level I loss, meaning vault prolapse**; if it stays, there is an isolated rectocele or enterocele.

POP-Q system

P1 *Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — POP-Q points table: hymen 0; Aa/Ap 3 cm (−3 to +3), Ba/Bp, C, D (absent after hysterectomy), gh, pb, tvl*

Not in the paper answer:

- Total vaginal length (tvl) is measured with the prolapse reduced, and the stage is set by the single most distal point.

Stage	POP-Q criteria	Baden–Walker grade (by eye)
0	Aa, Ba, Ap, Bp −3; C or D ≤ −(tvl − 2)	0: normal
I	Leading edge < −1 cm	1: upper vagina to midplane
II	Leading edge −1 to +1 cm	2: midplane to hymen
III	> +1 cm but < +(tvl − 2)	3: through/beyond hymen
IV	≥ +(tvl − 2): complete eversion	4: complete eversion

> **KEY POINT**
> Stage II is a 2-cm window around the hymen, and the III/IV boundary depends on tvl. Example: tvl 8, Bp +5 lies between +1 and +6, so **stage III**.

Other elements

- The **bulbocavernosus reflex** (S2–S4) is present in only about **70%** of normal women, so its absence alone is not pathological.
- For **prior mesh**, IUGA/ICS terms distinguish extrusion (through vaginal epithelium) from perforation (into bladder, urethra or another viscus).

4 · SUPPLEMENTAL EVALUATION

The uncomplicated patient needs few tests.

Urinalysis (UA) and post-void residual (PVR) are universal. For the AUA/SUFU 2023 index patient having SUI surgery, **UA, PVR and objective SUI demonstration** are required; cystoscopy, urodynamics (UDS) and imaging are not routine. In non-neurogenic OAB only UA is required (PVR optional).

P3 *SUI: evaluation in females, role of UDS, and surgical therapy in males — non-index features needing more tests: SUI not shown, neurogenic, urgency-predominant MUI, raised PVR, stage ≥3 POP without SUI, failed surgery; questionnaires; VALUE*

Not in the paper answer:

- Unexplained haematuria or pyuria also makes a patient non-index.

Symptom quantification

- A 3-day **bladder diary** is both diagnostic and therapeutic (see Ch 114), and its largest void equals functional capacity.
- Use the ICIQ-UI SF for UI and the **PFDI / PFIQ** for POP.
- **Pad tests** are mainly for research. In the ICS 1-hour test, 500 mL is drunk in <15 min, standard activities follow, and the pad is weighed.

	1-hour test	24-hour test
Normal	<1.4 g	<4.4 g
Mild	1–10 g	4–20 g
Moderate	11–50 g	21–74 g
Severe	>50 g	≥75 g

- In the **double-dye tampon test** (oral phenazopyridine and intravesical methylene blue), orange staining means a ureterovaginal and blue a vesicovaginal fistula.
- Measure **PVR** by bladder scan and repeat it. There is no agreed cut-off, but asymptomatic women are almost always <100 mL; a raised PVR changes counselling before SUI surgery or antimuscarinics.
- **UDS** are not needed in the index SUI patient (see Ch 107).
- **Cystoscopy** is not routine; use it for haematuria, urgency or irritative symptoms, recurrent UTI, prior sling or mesh, and pelvic radiation.

Imaging

Modality	Main indication
Voiding cystourethrogram	Reflux, diverticula, cystocele
Renal ultrasound	Hydronephrosis: chronic retention, **severe POP (ureteric kinking)**, fistula, neurogenic bladder
Translabial ultrasound	**Modality of choice for mesh** (CT and MRI poor); levator defects
MRI / dynamic MRI	**Urethral diverticulum (standard of care)**; complex or recurrent POP; symptoms out of proportion to findings

- A diverticulum gives the **3 Ds**: dysuria, postvoid dribbling and dyspareunia (see Ch 122).

5 · MANAGEMENT PATHWAY

ALGORITHM

1. History, examination, UA and PVR.
2. Exclude DIAPPERS causes, UTI, obstruction and tumour.
3. Classify the UI and grade the bother.
4. If there are red flags (pain, haematuria, recurrent UTI, pelvic mass, prior radiation or pelvic surgery, prolapse beyond the introitus, fistula), refer for specialist work-up.
5. If not, start initial management and reassess.
6. Add tests only if they change management; then surgery.

Diagnosis	Initial management
SUI	Lifestyle and behavioural therapy including pelvic floor muscle training (PFMT) (see Ch 114); † pessary; surgery if these fail
UUI / OAB	Lifestyle, behavioural therapy; antimuscarinic trial (see Ch 113)
Significant PVR	Treat constipation, review drugs; catheter drainage if PVR 200–500 mL, then reassess
Symptomatic POP	† Observation, PFMT, pessary or surgery, chosen by bother and goals

6 · LANDMARK TRIALS

Trial	Design	Result	Take-home
† VALUE	Randomised controlled trial, uncomplicated demonstrable SUI: office evaluation with or without UDS	Office evaluation alone non-inferior	Preoperative UDS not needed in index SUI

114 Conservative Management of Urinary Incontinence: Behavioural and Pelvic Floor Therapy, Urethral and Pelvic Devices

1 · DEFINITION AND PRINCIPLES

- **Conservative treatment** is any therapy that is neither surgery nor drugs.
- **Behavioural treatment** changes behaviour or environment, or teaches skills. It is effective and **essentially risk free**, so it is first-line in all guidelines for stress (SUI), urgency (UUI) and mixed (MUI) urinary incontinence, overactive bladder (OAB) and nocturia.
- Programmes are multicomponent and individualised, combining education, pelvic floor muscle training (PFMT), bladder training (BT), biofeedback (BF), electrical stimulation (E-stim) and lifestyle advice.
- Outcome is unpredictable, so offer it to anyone willing; dementia patients benefit from caregiver-led programmes.

	Outlet-directed	Bladder-directed
Target	Urethral closure	Bladder function via voiding habits
Examples	PFMT; behavioural training with urge suppression	BT; delayed voiding
	Patient-dependent	**Caregiver-dependent**
Examples	BT, PFMT, behavioural training	Timed voiding, habit training, prompted voiding
Needs	Intact function, learning, motivation	Knowledgeable, motivated caregiver
Suits	Cognitively and functionally intact	Impaired, nursing home

> **KEY POINT**
> Why first-line if surgery cures SUI more often? Conservative therapy is risk free and at least as effective as drugs. This asymmetry of harm drives every recommendation.

2 · ASSESSMENT

- A **bladder diary for 3–7 days** records intake, voids and volumes, urgency, leaks and pads. It estimates **functional capacity**, and its **self-monitoring effect** makes it assessment, outcome measure and treatment at once.
- **Education** comes first: anatomy and the mechanism of the patient's own leakage. Improvement is gradual and needs daily practice.

3 · LIFESTYLE MODIFICATIONS

Cheap adjuncts; evidence is weak except for weight loss.

Fluid management

- **Inadequate intake is more common** than excess; self-restriction causes dehydration, urinary tract infection (UTI), constipation and concentrated irritant urine.
- Excess intake worsens frequency, so reduce it if 24-h output is >~**2100 mL**.
- **Target** a **normal range, not restriction**: **30 mL/kg/day**, minimum **1500 mL**, at set times.
- Change the **timing**, not the total: stop fluids **3–4 h before bed** for nocturia. For oedema-driven nocturia, use stockings or leg elevation in the late afternoon, or a mid-to-late afternoon loop diuretic.

Caffeine and irritants

- Caffeine is a **diuretic and irritant** that raises filling detrusor pressure and is a risk factor for detrusor overactivity (DO).
- About **250 mg/day** (≈2 cups) is associated with moderate–severe UI in men.
- Only a subset is caffeine-sensitive. Try a 3–5-day reduction, tapering with decaf to avoid withdrawal headache.

Bowel function

- Constipation and straining are linked to lower urinary tract symptoms (LUTS) through shared innervation, cross-sensitisation and straining-induced **pudendal denervation**. **Faecal impaction** can cause OAB or overflow.
- Advise fibre **with adequate fluid**, exercise, and regular evacuation after meals.

Obesity, smoking, diet

- Obesity roughly doubles UI risk. **Loss of 5–10% body weight** sustained for 12 months substantially reduces UI, mainly SUI; bariatric surgery resolves UI in morbid obesity.
- Behavioural weight loss in overweight women is the **only grade A lifestyle recommendation**.
- For smoking, no UI recommendation is possible (grade D); advise cessation for general health. A fruit, vegetable and whole-grain diet helps prevent UI after menopause (B).

4 · PELVIC FLOOR MUSCLE TRAINING (PFMT)

Most failures come from a wrong contraction.

- PFMT, introduced by **Arnold Kegel in 1948**, is the cornerstone for SUI, UUI, MUI, urgency and frequency.
- **Mechanism**: contraction lifts the urethra and compresses it against the symphysis. Training causes striated muscle **hypertrophy**, which raises urethral pressure and improves bladder-neck support. There is also a **skill** route (the knack).

Assessment of PFM function

- A correct contraction gives **perineal elevation**; perineal descent means the patient is straining.
- On **digital vaginal exam**, levator ani lies just above the hymen at **4 and 8 o'clock**. Assess tone, symmetry, hold and quick flicks, and note **overactivity** (hypertonic, cannot relax).
- Scales: **Modified Oxford** is the most used; **Brink** scores pressure, duration and displacement.

Teaching

- Say: "Tighten as if stopping gas, pull up and in", without using buttocks or abdomen. For men: "move the penis up and down".
- The commonest errors are **co-contraction** (rectus, gluteals) and **Valsalva substitution**; counting aloud prevents breath-holding.
- **Verify correct contraction before prescribing.** Failure to isolate the muscle is probably the commonest cause of poor outcome; a pamphlet does not ensure it.

Regimen

No regimen is ideal; good results come with **45–60** contraction–relaxation pairs per day.

ALGORITHM

1. Empty the bladder.
2. **Lying**: 5 quick (1–2 s) and 5 sustained (5, building to 10 s) contractions.
3. Repeat **sitting**, then **standing** (total 30).
4. Rest as long as each contraction (work:rest **1:1 to 1:2**).
5. Do this twice daily = **60/day**.

- **The knack**: contract **just before and during** a cough, sneeze or lift. It reduces leakage immediately.

Evidence

- PFMT beats no treatment for SUI, UUI and MUI, short and long term; pure SUI does best.
- **Supervised training beats unsupervised** (grade A), and **age is no deterrent**.
- Antenatal PFMT in continent women prevents late-pregnancy and postpartum UI; give supervised postpartum PFMT if UI persists ~3 months.
- **After radical prostatectomy (RP)**, PFMT hastens continence in the first 3 months, and behavioural training helped men leaking years after RP. Adding E-stim or BF does not help.
- In **neurological OAB** (multiple sclerosis, incomplete spinal cord injury, Parkinson disease, stroke), PFMT is safe and first-line (B). PFMT adds to tolterodine for UUI.

Cones and devices

- **Weighted vaginal cones** (2 × 20 min/day) work by prompting a reflex contraction, but they **cannot be retained with POP or a wide introitus** and are no better than PFMT. Motion-sensing, app-guided devices can distinguish lift from strain and modestly improve results.

5 · URGE SUPPRESSION

- A well-timed PFM contraction **reflexly inhibits the detrusor**, whereas rushing raises bladder pressure.

ALGORITHM

1. When the urge comes, do not rush.
2. Stand still or sit, and relax.
3. Do several quick PFM squeezes, then distract yourself.
4. The urge subsides like a wave; then walk normally to void.
5. Once mastered, wait 5 min after the urge, then extend.

- Used alone for UUI, it gives a mean **60–80%** reduction in incontinence, and also reduces urgency, frequency and nocturia.
- For **nocturia**, suppress the urge on waking; void if it has not passed in 1–2 min.
- It is at least as good as antimuscarinics and adds to drugs. In men on an α-blocker, adding behavioural therapy reduced nocturia more than adding an antimuscarinic.

6 · BIOFEEDBACK (BF)

- BF is a **teaching technique**, not a treatment. It shows PFM activity in real time by manometry or surface electromyography (EMG); EMG separates PFM from abdominal activity, whereas pressure BF cannot separate lift from strain.
- BF **adds no clear benefit to supervised PFMT**. Use it for very weak PFMs, poor proprioception, an uncertain contraction, or motivation.

7 · ELECTRICAL STIMULATION (E-STIM)

- Non-implanted stimulation (vaginal or anal probe, or surface electrodes) produces **PFM contraction** and **detrusor inhibition**.

	Long-term (chronic)	Maximal
Intensity	Below sensory threshold	Just below pain threshold
Aim	Detrusor inhibition via pudendal afferents	Urethral closure (direct and reflex striated contraction)
Dose	6–12 h/day for months	15–30 min, several times a week to twice daily

Indication	Frequency
UUI	**5–20 Hz** (low)
SUI (including post-prostatectomy)	**20–50 Hz** (higher)
MUI	~20 Hz or alternating
All	Pulse 200–1000 µs, rectangular, biphasic

- E-stim beats sham but **adds nothing to PFMT**, so PFMT is first-line (A). Its **niche** is women who **cannot contract voluntarily**.
- **Tibial nerve stimulation** for UUI or OAB: transcutaneous equals percutaneous in efficacy, with fewer adverse effects (see Ch 115).

> **KEY POINT**
> E-stim is dosed by goal. Sub-sensory, low-frequency, prolonged stimulation inhibits the detrusor; near-painful, higher-frequency, brief stimulation contracts the sphincter. Volitional PFMT does both, so E-stim earns its place only where volitional contraction is absent.

8 · BLADDER TRAINING AND SCHEDULED VOIDING

P1 *Describe pathophysiology of OAB. Briefly outline management & describe nonmedical non-pharmacological management — BT breaks urgency, frequent voiding, reduced capacity cycle; void by clock not urge (dissociates voiding from urgency)*

Not in the paper answer:

- BT also works for SUI and MUI; it is grade A first-line in women, but evidence in men is insufficient.
- Adding behavioural training to drugs gives benefit that persists after the drug stops (UITN).

ALGORITHM

1. Start with a diary.
2. Set an initial interval of ~1 h while awake (30 min if needed).
3. Void on waking, at each interval and at bedtime, but not between.
4. If leaks and urgency are controlled, increase by **15–30 min/week**.
5. Target an interval of **3–4 h**; review weekly with reinforcement.

- BT beats no treatment for UUI, SUI and MUI, and the Agency for Healthcare Research and Quality (AHRQ) rates it equal to PFMT.
- **Delayed voiding** has no schedule: at each urge, suppress it, **wait 5 min**, and extend gradually. BT is triggered by the clock; delayed voiding by the urge plus a wait.

Caregiver-dependent toileting: elderly, nursing homes

Programme	Patient role	Key features
Timed voiding	Passive	Fixed 2–4 h by day; dryness goal. Also for impaired sensation and infrequent voiders. ~20% of frail elders become dry, 30–50% improve; a 2–3-day trial predicts success
Habit training	Passive	Schedule individualised from the diary, shorter than the usual interval, to **pre-empt** expected leaks; suits diuretic or time-patterned UI
Prompted voiding (PV)	Responsive	**Monitoring**, then **prompting**, then **praise**; every 2–2.5 h. 25–40% of nursing-home residents respond (daytime leaks ≤1); a 3-day trial predicts
Functional incidental training	Responsive	PV plus mobility and strength exercise
Bladder training	Active	Lengthens interval; needs ability and motivation

- **Only BT lengthens the interval**; caregiver programmes aim at dryness, and PV success depends more on the caregiver than the patient.
- First exclude reversible causes (DIAPPERS) and check post-void residual (PVR) (see Ch 105).

9 · HIGH-TONE PFM DYSFUNCTION

A tight pelvic floor needs relaxing, not strengthening.

- Raised tone causes pelvic pain, hesitancy and incomplete emptying. Treat by **down-training**: contract, then relax, with work:rest **1:4 or longer**; BF is useful.

10 · MECHANICAL DEVICES

Devices are **intravaginal** (support) or **intraurethral/meatal** (occlude); no device is proven superior.

Continence pessaries

- **Rationale**: the pessary **stabilises the bladder neck, lengthens the urethra and compresses it against the pubis**, like a sling. The knob of the ring sits at the urethrovesical junction.
- **Types** for SUI: incontinence **ring** (with knob), **dish** and **Hodge**. For pelvic organ prolapse (POP), support pessaries (ring, Gehrung) suit milder prolapse and space-filling ones (Gellhorn, cube, donut) suit advanced POP.
- **Fitting**: choose the largest comfortable size, one finger-breadth from the wall. The patient walks, coughs, strains and voids before leaving, with no expulsion, pain or obstruction.
- **Contraindications**: active pelvic infection, severe ulceration, silicone or latex allergy, **likely non-compliance with follow-up**.
- **Complications**: discharge and odour. Neglected use causes **erosion, impaction, vesicovaginal or rectovaginal fistula**, and bleeding.

- **Care**: follow-up every **24 weeks** is acceptable (versus 12; grade C), inspecting for erosion each time.
- About half improve. The International Consultation on Incontinence (ICI) says pessary or PFMT may be offered; pessary with PFMT beats PFMT alone for POP symptoms. See **ATLAS** under trials.

Other devices

- **Uresta** (reusable bell-shaped pessary) and **Impressa** (disposable tampon-like device) give suburethral support. Urethral inserts are balloon plugs for pure SUI, with risks of UTI, haematuria and migration.
- Use pads when treatment fails or is declined, and prevent **incontinence-associated dermatitis**.
- **An indwelling catheter is the last resort**: unmanageable retention, urine-contaminated sacral or perineal wounds, terminal care, or preference. Clean intermittent self-catheterisation (CISC) is preferred for retention.

11 · VAGINAL OESTROGEN

- For SUI with atrophy, PFMT with vaginal oestrogen beats oestrogen alone (C). Local oestrogen improves urgency, frequency, recurrent UTI and pessary tolerance (see Ch 113).

12 · ADHERENCE, PREVENTION AND DELIVERY

- **Adherence caps efficacy** and declines over time; realistic expectations, follow-up, reminders and linking exercise to daily cues help.
- **Prevention in childbearing**: supervised antenatal PFMT (12 weeks from 16–24 weeks' gestation) prevents late-pregnancy and early postpartum UI (A). Offer a postnatal programme after forceps or a baby ≥**4000 g**. Antenatal PFMT does **not** prevent postnatal prolapse.
- **Prevention at prostatectomy**: PFMT before surgery, with or without early postoperative PFMT, shortens time to continence.

13 · EVIDENCE GRADING: KEY ICI RECOMMENDATIONS

Grade (Oxford)	Meaning
A	Consistent level 1 evidence; effectively mandatory
B	Consistent level 2/3 studies or majority of randomised controlled trial (RCT) evidence
C	Level 4, or Delphi-processed expert opinion
D	**No recommendation possible** (not "weak")

ICI 2023 recommendation	Grade
Supervised PFMT first-line for UI in women of all ages	A
PFMT over E-stim as first-line	A
BT first-line for women with UI	A
Behavioural weight loss for overweight women with UI	A
Antenatal supervised PFMT in continent women to prevent UI	A
PFMT reduces POP symptoms (stage reduction uncertain)	A; B
PFMT over drugs first-line (fewer adverse effects, less dropout)	B
PFMT before surgery (surgery more effective but riskier)	B
Adding BF, cones or E-stim to PFMT: not routine	B/C
Fluid adequacy and caffeine reduction	B
Smoking cessation for UI	D

- AHRQ (2018): behavioural therapy is **more likely than anticholinergics to cure UUI** (high strength).

14 · LANDMARK TRIALS

Trial	Design	Result	Take-home
First behavioural vs drug RCT	Older women, UUI: behavioural vs titrated drug	Behavioural fewer leaks, more satisfaction	Behavioural at least equal to drugs for UUI
MOTIVE	Men with OAB, no obstruction: behavioural vs oxybutynin extended release (ER)	Equivalent reduction in voids	Behavioural therapy works in men
UITN (BE-DRI)	Urgency-predominant UI: tolterodine ER with or without behavioural	Combination better satisfaction, bother; persisted after drug stopped	Add behavioural training to drugs
ATLAS	Stress-predominant UI: pessary vs behavioural vs both	3 mo: behavioural better satisfaction; combination not better; no difference at 12 mo	Pessary reasonable alternative; behavioural first
Weight-loss RCT (PRIDE)	Overweight women: 6-month intensive weight loss vs education	Greater UI reduction	Recommend 5–10% weight loss (A)
GBT	Women with UI: one 2-h bladder-health class vs none	Modestly better ICIQ-SF score to 12 months	Group classes cheap, scalable

116 Retropubic Suspension Surgery for Incontinence in Females

1 · MECHANISM

- Stress urinary incontinence (SUI) arises from **urethral hypermobility** and **intrinsic sphincter deficiency (ISD)**, alone or combined.
- Suspension restores **pressure transmission** and a firm backstop for compression.

> **KEY POINT**
>
> Suspension corrects **hypermobility, not the sphincter**: it lifts and stabilises bladder neck and proximal urethra, so presupposes a mobile urethra. A fixed ISD urethra needs compression (sling, artificial urinary sphincter (AUS)).

2 · PATIENT SELECTION

- Do a stress test supine, then standing if negative; a Q-tip angle >30° means hypermobility.
- AUA/SUFU minimum: **post-void residual (PVR) and urinalysis**; urodynamics only if leak is not shown. Valsalva leak point pressure <60 or maximal urethral closure pressure <20 cm H_2O suggests ISD.
- **Indications** (Burch remains an AUA/SUFU option with midurethral sling (MUS), autologous sling and bulking): SUI with hypermobility, or the stress part of mixed incontinence (MUI); **concomitant abdominal surgery** (sacrocolpopexy); **difficult vaginal access**; **mesh aversion**.

Contraindications	Raise failure risk
Immobile vagina (cannot reach Cooper's)	Baseline urgency, preoperative anticholinergic
Immobile urethra (negative Q-tip ≈ **5-fold** failure): use sling or AUS	Obesity, menopause, age >60

Contraindications	Raise failure risk
ISD alone; "lead-pipe" (fixed, open) urethra	Prior hysterectomy or SUI surgery; advanced POP; MUI

3 · ANATOMY

- **Retzius** is an avascular preperitoneal space between symphysis and bladder. **Cooper's ligament** lies on the superior pubic ramus.
- The **arcus tendineus fascia pelvis (ATFP)** runs from symphysis to ischial spine.

4 · BURCH COLPOSUSPENSION

Paravaginal tissue is sutured to Cooper's; sutures elevate, and **fibrosis maintains** the repair.

ALGORITHM

1. Low lithotomy, prep including vagina, Foley.
2. Pfannenstiel or low midline incision.
3. Enter Retzius; sweep fat laterally.
4. Find bladder neck by Foley balloon and vaginal finger.
5. Dissect bladder off vagina **laterally**; expose Cooper's.
6. Place sutures, then cystoscopy.

P1 *Burch colposuspension — 2–4 sutures (0/1-0) each side, full thickness not lumen; distal at mid-urethra, proximal at bladder neck, lateral; complication rates*

Not in the paper answer:

- Tie **distal first** and **loosely** (Tanagho), leaving **≈ 2 fingerbreadths** between bladder neck and pubis; over-tightening obstructs.

KEY POINT

Danger points: never dissect midline over urethra or bladder neck (Santorini bleeding, injury, obstruction if sutures medial). Passing through Cooper's may bleed suddenly: use pressure, clips, sutures.

Approach	Bottom line
Open (Pfannenstiel)	Cure **85–90% at 1 y, ≈70% at 5 y**
Laparoscopic / robotic	Subjective cure equivalent to open to 18 months
Needle suspension (Pereyra, Stamey, Raz)	Abandoned: sutures tore through

5 · OTHER RETROPUBIC PROCEDURES

A higher anchor means more obstruction.

Technique	Anchor	Holds by	Obstruction
Burch	Cooper's ligament	Sutures; high V-shaped angle	Greater
Marshall–Marchetti–Krantz (MMK)	Symphyseal periosteum	Sutures; high	Greater
Paravaginal (Richardson)	ATFP	Scarring; flatter	Less
Vagino-obturator shelf (VOS)	Obturator fascia (or iliopectineal)	Scarring; flatter	Less

KEY POINT

AUS in women is second-line for ISD (recurrent or neurogenic). Place it at the bladder neck, preferably **abdominally**; transvaginal placement raises infection and erosion.

6 · PERIOPERATIVE CARE

- **Cystoscopy is mandatory**: no suture in bladder or urethra, and bilateral efflux. If efflux is absent, release sutures.
- For incomplete emptying, use clean intermittent catheterisation (CIC), preferred to suprapubic, until PVR is consistently **<150 mL**.

7 · OUTCOMES AND COMPLICATIONS

- Burch roughly equals retropubic MUS but is slower and causes more voiding dysfunction.
- Pubovaginal sling (PVS) is more effective but more obstructive (SISTEr).
- Add prophylactic Burch at sacrocolpopexy in stress-continent women (CARE); with existing SUI, MUS beats Burch.
- Voiding dysfunction (2–27%) is treated by CIC, then urethrolysis (Ch 118). De novo overactive bladder (OAB, 8–27%) needs obstruction excluded.
- **Posterior prolapse** (2.5–27%) follows because anterior elevation opens the cul-de-sac.
- **Osteitis pubis** is unique to MMK (0.7–2.5%) and causes suprapubic pain at 2–12 weeks; treat with rest and analgesia.

8 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SISTEr	Randomised trial (RCT), 655: Burch vs autologous fascial PVS	PVS higher cure at 2 and 5 y; more urinary infection, voiding dysfunction, de novo UUI	Sling more effective, more morbid
Ward–Hilton (TVT vs colposuspension)	First RCT, 348	No difference to 5 y; TVT more perforation; Burch more voiding dysfunction, later POP	MUS equals Burch, less invasive
CARE	RCT, 322 stress-continent, sacrocolpopexy with or without Burch	Less postoperative SUI with Burch, sustained 2 y	Add prophylactic Burch at sacrocolpopexy

118 Slings: Autologous, Biologic, Synthetic and Mid-Urethral

1 · HOW SLINGS WORK — THEORIES OF CONTINENCE

The theory decides where the sling sits and how tight.

P3 *Female pelvic floor, continence mechanism, classification and management of SUI — hammock (DeLancey) vs integral theory (Petros–Ulmsten: pubourethral ligaments, suburethral hammock, pubococcygeus); PVS at bladder neck, MUS loose at mid-urethra*

Not in the paper answer:

- Supporting integral theory, incontinent women have ≈**40% less collagen** in skin and round ligament MCQ .
- **Urethral hypermobility is a SIGN** of damaged support, **not the cause** of stress urinary incontinence (SUI).
- A pubovaginal sling (PVS) sits under **minimal tension at the bladder neck** and rebuilds the hammock. A mid-urethral sling (MUS) lies **loose under the mid-urethra** and arrests the posterior wall. Neither should obstruct at rest.

Historical milestones MCQ

- McGuire reintroduced the fascial PVS (1978). The tension-free vaginal tape (**TVT**, 1995) was the first retropubic MUS and defined the two complications **unique to MUS**: urinary tract mesh penetration and vaginal exposure. The transobturator route followed in 2001 and the single-incision mini-sling (SIMS) in 2006.

Guideline status (AUA/SUFU)

- MUS for the index patient is a **strong recommendation**, grade A. Either route is conditional, grade A; SIMS is conditional, grade B.
- Mesh-specific risks **and** non-mesh alternatives must be discussed before a synthetic MUS (Clinical Principle).

2 · PREOPERATIVE ASSESSMENT AND PATIENT SELECTION

P3 *SUI: evaluation in females, role of UDS, and surgical therapy in males — AUA/SUFU minimum evaluation (history, exam, cough stress test supine then standing, PVR, urinalysis); indications for UDS;*

ValUE; ISD cut-offs

Not in the paper answer:

- Q-tip movement **>30°** on straining defines hypermobility.
- Further evaluation, including urodynamic studies (UDS), is needed for unclear diagnosis, neurogenic bladder, urgency-predominant mixed urinary incontinence (MUI), high post-void residual (PVR), POP-Q stage ≥3 without SUI on reduction, or prior surgery. Cystoscopy is not part of the index evaluation.

Urodynamic studies — the controversy

- UDS may be **omitted** in the index patient with demonstrable SUI (conditional, grade B), based on **ValUE**.
- A maximal urethral closure pressure (MUCP) **<20 cm H_2O** defines type 3 SUI, that is intrinsic sphincter deficiency (**ISD**). A Valsalva leak point pressure (VLPP) **<60 cm H_2O** also means ISD; a value **>90** means little or no ISD.
- ISD is a **matter of degree** and now largely a clinical diagnosis; it matters most with a **hypomobile urethra**.
- **Detrusor overactivity (DO)** is the most consequential bladder finding and may improve or worsen, so consider treating it first. Detrusor underactivity predicts voiding difficulty but should **not exclude** surgery.
- Offer non-surgical options to all: weight loss, **pelvic floor muscle training (PFMT)** before surgery for most, and pessaries.

3 · CHOOSING THE SLING

Situation	Preferred option	Reasoning
Index SUI, mobile urethra	MUS (retropubic or transobturator); SIMS acceptable	Equal efficacy, least morbidity
ISD with mobile urethra	Retropubic preferred over transobturator	Meta-analyses favour retropubic
Fixed, immobile urethra	Retropubic MUS, autologous PVS, bulking, adjustable sling	AUA Expert Opinion; mobility predicts MUS success
Recurrent SUI after failed MUS	Retropubic MUS or autologous PVS	Retropubic better than transobturator for recurrence
Mesh complication, urethral injury, fistula, diverticulum, reconstruction	**Autologous PVS**	No mesh near a damaged urethra; adds tissue

Situation	Preferred option	Reasoning
Declines mesh	Autologous PVS or Burch see Ch 116	—
Neurogenic SUI on clean intermittent catheterisation (CIC)	Autologous PVS (may be tight)	Only once UDS confirm safe capacity and compliance
Concomitant pelvic organ prolapse (POP) surgery	MUS, PVS or Burch	Conditional, grade C

4 · PUBOVAGINAL SLING

- **Indications:** primary or recurrent SUI from **hypermobility and ISD**, stress-predominant MUI, anteroapical POP, neurogenic SUI, **urethral reconstruction** (trauma, mesh perforation, interposition at fistula or diverticulum repair), and failed Burch or MUS.
- **Autologous fascial PVS is the gold standard for all forms of SUI.** It gives **dynamic compression without obstruction**; too much tension causes retention and erosion.
- Width **<2 cm** compresses without a constricting band.

Material	Behaviour	Status
Autologous — rectus fascia, fascia lata	Best ingrowth and neovascularisation, minimal reaction; negligible erosion	**Gold standard**; cost is harvest pain and wound complications
Allograft	Processing weakens it; variable, less durable	Mostly withdrawn
Xenograft (porcine, small intestinal submucosa (SIS), bovine)	Weaker, encapsulated; cure falls with time; porcine inferior in trials	Mostly withdrawn
Synthetic at bladder neck	Urethral perforation **15×** and vaginal exposure **14×** other materials (especially woven polyester)	**No longer used at bladder neck**
Vaginal wall sling	Weak; inclusion cysts, vaginal shortening	Historical

KEY POINT

Fibrosis paradox: more fibrosis should mean better support, but the inflammation that drives it also degrades the sling and causes erosion. Autologous fascia remodels with minimal inflammation, the ideal balance.

- **Rectus or fascia lata:** strength and cure are equal. Fascia lata avoids abdominal wound problems but causes thigh pain; use it after **ventral hernia repair**.

5 · AUTOLOGOUS PVS — OPERATIVE PROCEDURE

- **Counsel** about voiding dysfunction and de novo urgency, and teach CIC in selected women. Give one dose of cefazolin and mechanical thromboprophylaxis.

ALGORITHM

1. Through a 6–7 cm Pfannenstiel incision, mark and excise a **2 × 8 cm** rectus graft.
2. Close the fascia with No. 1 polydioxanone (PDS).
3. Clean the graft, stitch No. 1 PDS to each end and keep it in saline.

- Keep the transverse incision **≥2 cm above the symphysis** for tension-free closure. A fascia lata graft is taken over the iliotibial band, and the fascia lata is not closed.
- **Vaginal part:** make an **inverted-U incision**. With the **bladder empty**, perforate the **endopelvic fascia** and finger-dissect the retropubic space.

ALGORITHM

1. With the bladder empty, pass Stamey needles **from above**, on the back of the pubis, into the vaginal incision.
2. Perform **cystoscopy (70° lens)**. If there is an injury, repass **more laterally** and re-scope after each pass.
3. Confirm ureteric efflux, centre the graft and tag its ends above.

- Do any POP repair **before tensioning** and remove the weighted speculum.
- **Tension:** tie the sutures over the rectus with **two fingerbreadths** of slack, deliberately **loose**. Tie tighter only if lifelong CIC is planned.
- Postoperatively, do a voiding trial on day 1; if it fails, use Foley or CIC and retry within 5 days. No heavy lifting and no intercourse for **6 weeks**.

6 · PVS OUTCOMES

PVS rarely fails for stress leakage; "failure" is usually urgency.

- Results are durable even in complex patients (radiation, spinal cord injury, trauma). Cure stays high even with MUCP <4 cm H_2O, so **severe ISD is not a contraindication**.
- **Recurrent SUI:** a repeat PVS, or a PVS after failed MUS, succeeds in most.
- The commonest "failure" is **storage symptoms or urgency urinary incontinence (UUI)**; de novo UUI occurs in 2–22%, predicted by prior POP surgery.
- **Cochrane, PVS versus MUS:** PVS is probably no better, maybe less effective; MUS is shorter, with fewer complications (except bladder perforation) and less voiding and storage dysfunction.
- A damaged urethra does worst, yet PVS still dries most. In **MUI**, PVS cure equals that of pure SUI but the urgency outcome is unpredictable.

7 · PVS COMPLICATIONS AND MANAGEMENT

Voiding dysfunction and bladder outlet obstruction (BOO)

- Incidence is 2.5–35%, **more than Burch** (SISTEr: all outlet procedures were in the PVS arm). Retention >4 weeks is ≈8%; **permanent retention is ≤5%**.
- The usual cause is **excess tension**; poor contractility and **straining to void** add to it.
- It presents as weak stream, retention, **storage symptoms only**, recurrent urinary tract infection (UTI) or pain, even with a low PVR. Exclude new POP kinking the urethra.
- Prediction is unreliable, so a low detrusor pressure means **counselling about CIC**, not refusing surgery. Evaluate by examination, PVR, cystoscopy and selective video-UDS; a high-pressure, low-flow pattern is often **absent in women**.

KEY POINT

The key diagnostic criterion is the temporal relationship between surgery and symptoms, not a urodynamic number. Preoperative UDS does not predict obstruction (SISTEr).

ALGORITHM

1. Most transient retention resolves within **10 days**: use catheter or CIC, timed voiding, PFMT, anticholinergic.
2. Autologous obstruction often improves, so wait up to **3 months**.
3. If retention is complete, act earlier: at **<6 weeks**, loosen the sling under anaesthesia by downward urethral traction.
4. If **>6 weeks** or conservative care fails, then do sling incision or urethrolysis.

- Downward loosening is **not for synthetic slings**. Dilation risks fibrosis, and **transurethral bladder-neck incision fails** because the sling is extraluminal.

Procedure	Features
Sling incision (transvaginal)	84–100% success; least morbid, so **first choice**; ≈20% recurrent SUI, not prevented by interposition
Transvaginal urethrolysis	Least effective for PVS: cannot relieve suburethral compression
Suprameatal urethrolysis	Better access to the lateral wings
Retropubic urethrolysis	Best results; after failed transvaginal attempt

- Urethrolysis succeeds in 65–93%, with recurrent SUI in 0–19%. Failure is usually **inadequate lysis**, and a repeat relieves ≈90%.
- **Storage symptoms persist** in over half: use an antimuscarinic or β3-agonist, **sacral neuromodulation (SNM)**, or onabotulinumtoxinA 100 U.
- Recurrent SUI after lysis is treated with bulking, repeat PVS or MUS.

Perforation, exposure and other complications

- Perforation is rare with fascia. Urethral perforation needs **sling excision and repair** (with or without Martius flap); bladder erosion is removed endoscopically.
- An exposed synthetic PVS is removed, and recurrent SUI is treated with a second, **autologous** PVS.
- Mortality is ≈**3 per 10,000** anti-incontinence procedures; harvest-site **incisional hernia** occurs in ≈1%.

8 · MID-URETHRAL SLING — MECHANISM AND MATERIALS

- The MUS **secures the posterior urethral wall**, with dynamic **kinking**. Mesh is fixed along its **whole length**, so 50–93% stay dry after midline incision.
- Preoperative **immobility predicts failure**; the ideal is a loose sling with a mobile urethra. **Exception, SIMS:** success needs restricted mobility, so it is set tighter.

> **KEY POINT**
>
> **Why does a loose tape work?** At rest it does not touch the urethra, so voiding is unobstructed. On coughing the mobile urethra descends onto the tape; the posterior wall is arrested, the urethra kinks and closes. A fixed urethra never meets the tape, and tightening the tape to compensate obstructs.

- The material is **type 1 (Amid) macroporous monofilament polypropylene** with pores **≥75 µm**, which admits macrophages and fibrovascular ingrowth. Microporous tapes (<50 µm) had high exposure and were withdrawn.

9 · MUS OPERATIVE PROCEDURES

Each route has its own danger structures.

P2 P1 *Mid-urethral slings for SUI: TOT, TVT and TOT versus TVT; TOT in SUI — TVT bottom-up steps,*

TOT outside-in/inside-out steps, tensioning, Cochrane complication rates

Not in the paper answer:

- Document counselling about the **permanent implant**, mesh-specific complications and non-mesh alternatives.
- Bottom-up beats top-down, with better subjective cure and fewer perforations and exposures.

ALGORITHM

1. Infiltrate local; make two suprapubic stabs ≈2 cm from midline.
2. Make a 1.5 cm midline vaginal incision 1.5 cm from the meatus.
3. Make minimal lateral tunnels to the endopelvic fascia, **not perforating** it (unlike PVS).
4. The catheter guide deflects the bladder to the opposite side.
5. Pass the trocar through Retzius and rectus, **hugging the back of the pubis**; repeat on the other side.
6. Perform cystoscopy.
7. Tension loosely (clamp between tape and urethra), remove sheath, trim.

- Keep the trocar in **metal-to-bone contact** with the back of the pubis, so it stays extravesical and extraperitoneal.
- **Transobturator, outside-in (TOT):** hips **hyperflexed (120°)**; puncture the labiocrural fold at the adductor longus notch, pierce the obturator membrane and turn around the ramus onto the finger. Check the vaginal sulcus for a button-hole.
- **Inside-out (TVT-O)** has the same cure, with more voiding dysfunction and fewer vaginal perforations.
- **SIMS** uses only a vaginal incision and anchors in **obturator internus**; set it **tighter**, close to the urethra.

Feature	Retropubic (TVT)	Transobturator	SIMS
Path / anchor	Retzius, rectus fascia; U-shape	Obturator foramen; flatter hammock	Obturator internus only
Cure (index SUI)	Equal to transobturator up to 5 y; best long-term data (>17 y)	Equal; slightly more repeat surgery	Non-inferior to transobturator (TVT-Secur excluded)
ISD / recurrent SUI	**Preferred**	Worse with low VLPP	Limited data
Bladder perforation	**4.5%**	**0.6%**	Rare
Vascular/visceral injury, bleeding, voiding dysfunction	More	Less	Less
Pain	Suprapubic 2.9%	**Groin 6.4%** vs 1.3% retropubic	Least
Vaginal exposure	About 2%	About 2%	Similar (excluding TVT-Secur)
Cystoscopy	Mandatory (70°)	Recommended; check the urethra	Essential (30° suffices)

10 · INTRAOPERATIVE CYSTOSCOPY

P1 *Management of perforated mid urethral sling — 70° lens, full bladder; bladder injury: repass laterally and continue; urethral injury: no mesh, staged repair*

Not in the paper answer:

- Inspect the urethra too after transobturator and SIMS passes. A bladder injury is benign: leave a catheter overnight to 72 h.

11 · MUS OUTCOMES

These are the figures quoted when counselling.

- Retropubic slings are the most studied (17–20 years); objective cure stays high and subjective cure is lower. Cochrane (81 trials): retropubic and transobturator cure are equal from short to long term.
- TVT equals colposuspension at 5 years, with less de novo urgency and less later POP.

- In TOMUS at 5 years, satisfaction stayed high (79% retropubic, 85% transobturator) despite falling objective success.
- Long term, **de novo urgency rises** (≈⅓ at 10–20 years); retention is 0–7%.
- In Cochrane, SIMS did worse than retropubic, driven by **TVT-Secur (withdrawn)**; others about equal transobturator, with less pain. The key long-term endpoint is **reoperation**.

Group	Key points
MUI	Urgency resolves in ≈60–75% (best when stress-predominant). De novo urgency risk rises with age, obesity, nocturia, DO and low capacity. Persistent DO: transobturator < retropubic < PVS. New storage lower urinary tract symptoms (LUTS) may mean BOO
ISD	About 50% cure with urodynamic ISD. An immobile urethra predicts failure more strongly than a low VLPP, and low VLPP is not a contraindication to retropubic MUS. Transobturator: 12× odds of persistent SUI with low VLPP
Recurrent SUI	No level 1 evidence; repeat MUS is commonest, retropubic better than transobturator. Cure ≈81–90% but below primary surgery (≈62% vs 86%); more urgency, and more bladder perforation after prior retropubic surgery
Concomitant POP	A concomitant sling lowers postoperative SUI and reoperation but raises voiding dysfunction. About 30% of SUI resolves after POP repair alone. A prophylactic sling only after counselling (OPUS)
Elderly	Safe; similar or lower cure, more urgency and voiding dysfunction. Q-tip >30°: 90% continent; <30°: 42%; <10°: 80% stay incontinent. Cure ≈64% at ≥75 y vs 89% at 55–64 y. **Frailty, not age**, predicts readmission. Do not exclude by age (AUA Expert Opinion)
Obese	Success >80% but lower (≈87% vs 66% at 5 y), worse with BMI >35; more wound infection, fewer bladder punctures

12 · MUS COMPLICATIONS AND MANAGEMENT

> **KEY POINT**
>
> **Concurrent or staged sling at POP repair**. Concurrent: one operation and less SUI (≈39% falls to 8–19%), but more obstruction, storage symptoms and mesh events. Staged: avoids an unnecessary sling but commits a third to a second operation. Decide after testing for occult SUI with the prolapse reduced.

- Bladder trocar injury is ≈2–4%, almost all retropubic; risk rises with prior retropubic surgery and thin habitus. Vascular (≈0.07%) and bowel (≈0.04%) injuries are rare, but deaths have been reported.
- †Unexplained hypotension, falling haemoglobin, pain, ileus or sepsis after a retropubic sling needs urgent imaging or exploration.
- Bleeding occurs in 2–3%, more with retropubic slings. A retropubic haematoma is usually **asymptomatic, so observe**.
- **UTI is the commonest complication** (3–4% at 30 days). It is less after transobturator slings and marks a worse course.
- Vaginal wound infection is 0.7–2%; rarely there is abscess or **adductor myositis**, which may present years later.
- **Groin pain** after transobturator slings affects 16–26% early and ≈4% at 12 months. Treat with NSAIDs, physiotherapy and trigger-point injection; if refractory, excise, and **complete removal** relieves more than partial.

Voiding dysfunction and retention

- The tape is **too tight** or **too proximal**, or POP kinks the urethra. Obstruction is commoner after retropubic slings (TOMUS: 2.7% vs 0%). Evaluate by **temporal relation**, examination, PVR, cystoscopy and selective UDS.

ALGORITHM

1. Retention or high PVR: CIC or catheter; most resolve within 2 weeks.
2. If obstruction persists, perform a **transvaginal midline sling incision at 2–4 weeks**.
3. Beyond 3 months the tape is fixed, so excise its lateral portion.
4. Formal urethrolysis is rarely needed.

- **Urethral dilation is harmful; do not perform it**. After incision 70–100% stay continent. **Early** release gives better voiding and **less** recurrent SUI.

KEY POINT

Why is the timing different from PVS? Fascial obstruction often settles, so one waits up to 3 months. Mesh is fixed by ingrowth and rarely loosens; late release needs wider dissection and causes more recurrent SUI, so release within ≈4 weeks.

Storage symptoms, exposure and sexual function

- De novo overactive bladder (OAB) is ≈6–12%, equal for all MUS types. A clear temporal link means **partial BOO first**, and **cystoscopy is mandatory**. Treat with behavioural measures and drugs, then sling revision, SNM, percutaneous tibial nerve stimulation (PTNS) or botulinum toxin.
- Vaginal exposure is ≈2%; †it causes discharge, bleeding, dyspareunia or partner discomfort. †Small exposures get topical oestrogen with or without excision; larger ones are excised. Urinary tract mesh needs excision and repair, with no immediate new mesh see Ch 119.
- De novo dyspareunia is ≈3–14%, and sling removal may relieve it. Many women improve as **coital incontinence** stops.

13 · ADJUSTABLE SLINGS

- **Remeex** (mesh on sutures to a suprapubic varitensor, adjusted after surgery) is a niche option for **complex SUI**. Cure is ≈85–90%, but complications reach 32–51% and there is no level 1 evidence.

14 · ASSESSING THE FAILED OR TROUBLESOME SLING

ALGORITHM

1. Define the symptom: SUI, storage, voiding, or pain/discharge.
2. Establish the **timeline** in relation to surgery.
3. Examine for exposure, POP, urethral mobility and tenderness over the tape.
4. Check PVR and urine culture, perform **cystoscopy**, and add UDS selectively.
5. Tailor treatment to the cause.

- If SUI recurs with a mobile urethra, repeat a retropubic MUS or do a PVS; if the urethra is fixed or there is ISD, use PVS, bulking or an adjustable sling.
- If storage symptoms have a temporal link, treat obstruction first. Pain or exposure needs excision (complete for pain) see Ch 119.

15 · MESH CONTROVERSY AND REGULATORY STATUS

- The **2010 FDA** safety communication on mesh for POP and SUI was based on **MAUDE** database reports. Professional bodies still endorse MUS with documented counselling.
- †The 2019 FDA order stopping transvaginal mesh concerned **POP**, not MUS. The UK (2018) paused MUS use pending safeguards see Ch 119.

16 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ValUE	RCT, non-inferiority, 630, uncomplicated SUI: UDS vs office evaluation	Office evaluation non-inferior at 1 y	UDS may be omitted in the index patient
SISTEr	RCT, 655: rectus PVS vs Burch	PVS higher cure (47% vs 38% at 2 y); more voiding dysfunction, UTI and UUI; all BOO surgery in PVS arm; persisted at 5 y	PVS more effective but more obstructive
Ward–Hilton	RCT: TVT vs open colposuspension	Equal to 5 y; TVT more bladder injury; Burch more morbidity and POP	MUS replaced Burch
TOMUS	RCT, 597: retropubic vs transobturator	80.8% vs 77.7% (1 y, equivalent); surgery for voiding dysfunction 2.7% vs 0%; †more groin and neurological symptoms with transobturator; 5 y 51% vs 43% (equivalence not met)	Either; retropubic slightly more durable, more voiding dysfunction
OPUS	RCT (327): POP repair with or without prophylactic retropubic MUS in stress-continent women	Incontinence at 12 months 27% vs 43%; NNT 6.3; more perforation, UTI and bleeding	Reduces SUI but adds risk, so counsel

117 Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse

1 · GOALS AND PRINCIPLES

Prolapse is a quality-of-life problem, so operate for bother.

P2 *Vaginal prolapse repair — QoL condition, first operation best chance, restorative/compensatory/obliterative, vaginal (less morbid) vs abdominal (more durable), apex is the cornerstone*

Not in the paper answer:

- **Success means absence of symptoms** (no bulge), which drives satisfaction more than anatomy does.
- Obliterative surgery is contraindicated if intercourse is desired. Open and laparoscopic routes have equal complications, but laparoscopic recovery is faster.
- **The first operation is the best chance**, because planes are then lost and recurrence rises with each re-attempt. Preoperative education improves satisfaction, and declining surgery is a legitimate choice.

2 · CHOICE OF PROCEDURE BY COMPARTMENT

P1 *Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — POP-Q points (A distal, B proximal, D only with cervix), DeLancey levels, operation by defect, SSLF and*

promontory anatomy, uterus-preserving options

Not in the paper answer:

- Enterocele (point D) is closed by culdoplasty: **McCall** vaginally, **Halban** or **Moschcowitz** abdominally.

Compartment	POP-Q point	Organ
Anterior	Aa / Ba	Urethrocele / cystocele

Compartment	POP-Q point	Organ
Middle	C / D	Cervix or cuff / enterocele
Posterior	Ap / Bp	Rectocele / enterocele; perineal body

Site	Vaginal	Abdominal
Aa — urethra	Anterior repair; bladder neck suspension; sling	Retropubic urethropexy
Ba — bladder	Anterior repair; paravaginal; colpocleisis; wedge colpectomy	Paravaginal; sacrocolpopexy
C — cervix/cuff	Uterosacral ligament suspension (USLS); iliococcygeus; sacrospinous ligament fixation (SSLF); Manchester; hysteropexy; vaginal hysterectomy; colpocleisis	Hysterectomy; USLS; sacrocolpopexy; uterine suspension
D — cul-de-sac	**McCall** culdoplasty	**Halban**, **Moschcowitz** culdoplasty
Ap — posterior	Posterior repair; site-specific repair	Colpoperineopexy

KEY POINT

Only paravaginal repair and uterosacral suspension are done by both routes. McCall is vaginal; Halban (linear sutures) and Moschcowitz (purse-string, which may angulate the ureter) are abdominal.

3 · ANTERIOR COMPARTMENT

Anterior repair has the highest recurrence.

- A defect may be **central** (midline thinning of the pubocervical fascia) or **lateral/paravaginal** (detached from the arcus tendineus fasciae pelvis, ATFP); most are combined. **Anterior repair does not suspend the apex.**

ALGORITHM

1. In lithotomy, catheterise, hydrodissect and incise in the midline from apex to bladder neck.
2. Dissect the epithelium off the pubocervical fascia laterally to the ATFP and reduce the bladder.
3. Plicate with interrupted 2-0/3-0 delayed-absorbable sutures from bladder neck to apex.
4. Trim judiciously and close; give intravenous (IV) dye, do **cystoscopy** and pack.

- Deep sutures may enter or kink the bladder or ureter, so completion cystoscopy is mandatory.
- Anterior repair is abandoned for stress urinary incontinence (SUI): AUA meta-analysis failure approached **40%**.
- Recurrence depends on the **definition**: 30–46% for "POP-Q ≥2" versus 9–23% for "beyond hymen".
- Polypropylene lowers recurrence versus native tissue but causes far more graft complications without better QoL. Hence the **withdrawal of transvaginal synthetic POP mesh** (US 2019) see Ch 119.
- A concomitant sling **halves anterior recurrence**, more than any graft.

Complication	Point / management
De novo SUI	Rises with stage: stage 2 **41%** , early 3 **53%** , advanced 3–4 **66%** ; point Ba predicts it
Detrusor overactivity	De novo in 5–7%; pre-existing overactivity resolves in up to 63%
Retention	Usually from the sling: clean intermittent catheterisation (CIC); if persistent, sling incision or urethrolysis
Bleeding (>350 mL)	Bleeding means the wrong plane; worse at redo

Complication	Point / management
Bladder injury	Drain the bladder before perforating endopelvic fascia; two-layer closure; Martius if irradiated
Others	Ureteric obstruction, vesicovaginal fistula (VVF), vaginal shortening, dyspareunia

> **KEY POINT**
>
> **Confirming ureteric efflux**: indigo carmine (blue, IV), fluorescein 10% (green, IV) or phenazopyridine (orange, oral). If there is no efflux, pass a ureteric catheter or remove plication sutures. Never close without proving patency.

Paravaginal repair

- Reattach the pubocervical fascia to the ATFP with non-absorbable sutures 1 cm apart (**5–7** vaginal, **4–6** abdominal), then do cystoscopy.
- Lateral recurrence is 2–8%, but central recurrence is ~**22%**, so these are independent failures. The abdominal route is more durable.
- **Neither colporrhaphy nor paravaginal repair treats SUI.** The vaginal route has more serious complications (transfusion 12%).

4 · OCCULT SUI AND THE CONCOMITANT SLING

- High-stage anterior POP kinks the urethra and masks SUI; SUI on prolapse reduction occurs in 8–66% of continent women.
- Test **with prolapse reduced** (pack, speculum or pessary); sensitivity is low and specificity high.
- **Staged** argument: sling obstruction needing intervention (8.5%) equals intervention for SUI without a sling (8.3%).
- The **OPUS calculator** uses 7 predictors (accuracy 0.72 vs 0.54 for the stress test). In **CUPIDO**, urodynamics added nothing in asymptomatic women.

> **KEY POINT**
>
> **A sling also helps the prolapse**: distal anterior support makes the cystocele repair last. **Why not everyone?** In OPUS it caused more bladder perforation, UTI, bleeding and incomplete emptying. Weigh per patient.

ALGORITHM

1. In advanced anterior POP, do a reduction stress test, with or without urodynamic studies (UDS).
2. If SUI is symptomatic, occult or urodynamic, then offer a concomitant procedure.
3. If she is continent and the test is negative, then weigh individual risk and benefit (OPUS calculator).
4. The sling is **selective, not routine**.

5 · APICAL COMPARTMENT

The apex is the cornerstone of prolapse repair.

Uterosacral ligament suspension (USLS)

- **The ureter lies 4 cm from the ligament at the sacrum but only 0.9 cm at the cervix**. Injury is usually **kinking**, not ligation. Sutures near the sacrum can entrap S1–S4 roots, causing buttock and posterior thigh pain.
- The **safe zone** is the intermediate portion, ~1 cm posterior, at the level of the ischial spine; pass needles **lateral to medial**.

ALGORITHM

1. After vaginal hysterectomy, place cuff stays at 3 and 9 o'clock, excise the enterocele sac, pack bowel and palpate the spines.
2. Place Allis clamps at 5 and 7 o'clock and pass **2–4** non-absorbable sutures per side.
3. Do any anterior repair or sling now.
4. Bring the sutures through the apical pubocervical and rectovaginal fascia and tie.
5. Perform **cystoscopy after tying, before trimming**. If there is no efflux, remove the most lateral ipsilateral suture first.

- Compared with SSLF, USLS keeps the midline axis, closes the cul-de-sac and risks the pudendal vessels less.
- Apical success is ~98% and anterior ~81%; ureteric kinking or injury occurs in **1–11%**.
- **OPTIMAL**: USLS equals SSLF (~65% at 2 years on a strict composite), and perioperative pelvic floor muscle training (PFMT) adds nothing.

Sacrospinous ligament fixation (SSLF)

- SSLF is extraperitoneal and preserves length. The ligament lies under coccygeus; the pudendal bundle runs round the spine and the inferior gluteal vessels lie behind.
- The **safe zone** is **1.5–3.0 cm medial to the ischial spine**, at limited depth. Too lateral risks the pudendal bundle; too medial hits weak ligament and sacral nerves.

ALGORITHM

1. Mark the apex, incise in the midline, enter the perirectal space (rectum medial) and palpate spine and ligament.
2. Place retractors and pass 2 sutures (Capio or Miya hook).
3. Complete other repairs, then attach the sutures to the apex (pulley stitch if permanent).
4. Close the vagina halfway and tie the apex flush, with **no suture bridge**.
5. Do cystoscopy and close.

- SSLF is usually right-sided, with 2 sutures 2 cm apart. The **pulley stitch** keeps the knot out of the lumen.
- Cure is ~85% and apical recurrence ~5%; stage 4 predicts recurrence.
- **Drawback:** the axis is deflected posteriorly and to the right, causing **anterior recurrence** (~18%).
- Gluteal pain occurs in **15%** and resolves in 2–3 months. Pudendal injury, rectal injury and haemorrhage can also occur.

KEY POINT

Both ligaments are safest in their **middle third** and dangerous at their bony ends: the intermediate portion for USLS, and 1.5–3 cm medial to the spine for SSLF.

Iliococcygeus suspension is for deficient uterosacrals, with sutures **1 cm distal to the ischial spine**; anterior recurrence is commonest.

Sacrocolpopexy (SCP)

- **Indications:** other defects (including SUI) repairable at one sitting, failed vaginal repair, young sexually active women, isolated vault prolapse, or wish for the most durable repair. It maximises vaginal length without distortion.
- Use **type I macroporous monofilament polypropylene**; fascia lata recurs in ~1/3 versus <1/10 with polypropylene.

ALGORITHM

1. Low lithotomy with a vaginal stent or EEA (end-to-end anastomosis) sizer; sigmoid to the left.
2. Incise the peritoneum over the promontory and along the **right** of the rectum (watch the right ureter) down to the anterior longitudinal ligament.
3. If there is a large enterocele, then add a Halban culdoplasty.
4. Dissect the bladder off the anterior vagina (6–8 cm) and the rectum off the posterior wall.
5. Attach a Y-mesh to both walls **without entering the lumen**, tension it, and fix it with 2–3 permanent sutures at **S1**.
6. **Retroperitonealise** the mesh, add an anti-incontinence procedure if indicated, and perform cystoscopy.

- Fixation too low (S3–4) risks the presacral venous plexus; too high risks the middle sacral vessels and the **left common iliac vein**.
- Avoid hypersuspension, which causes pain and de novo SUI. Sutures or tacks into the promontory can cause spondylodiscitis.
- Open, laparoscopic and robotic SCP succeed in 87–100%; the robot saves little blood but costs more.
- Failures are mainly **posterior or distal**, with almost no apical retreatment; ~20% later need vaginal repair, so counsel.

SCP complication	Rate / management
Haemorrhage (presacral)	~4%; †pressure, sutures, bone wax, sterile thumbtacks
Cystotomy / enterotomy / ureter	3% / 1.6% / 1%; bladder is the commonest robotic visceral injury
Ileus / small bowel obstruction	3.6% / 1.1%
Mesh exposure	**~10% by 7 years** (E-CARE); more with microporous mesh; excision see Ch 119

- **Hysterectomy at SCP:** total hysterectomy raises exposure, so go **supracervical** or close the cuff meticulously.
- Cochrane, vaginal apical repair vs SCP: the vaginal route gives more prolapse awareness (relative risk 2.1), repeat surgery (2.3), recurrence (1.9), SUI (1.9) and dyspareunia (2.5), yet it suits many older women.

ALGORITHM

1. If she is elderly or comorbid and not sexually active, then offer colpocleisis.
2. If she is fit, has primary prolapse and prefers the vaginal route, then offer USLS or SSLF, with or without vaginal hysterectomy or hysteropexy.
3. If young or sexually active, or with recurrence, short vagina or high stage, then offer SCP (minimally invasive).

6 · HYSTERECTOMY VS UTERINE PRESERVATION

- †Hysterectomy alone does not cure prolapse, so always resuspend the apex.
- Sacrospinous hysteropexy is non-inferior to vaginal hysterectomy with USLS.
- †Preservation needs a normal uterus: exclude abnormal bleeding, endometrial or cervical disease and Lynch syndrome, and keep cervical screening up to date.

7 · OBLITERATIVE SURGERY — COLPOCLEISIS

For frail older women it is quick, low-risk and effective.

- **Total colpocleisis** (after hysterectomy) denudes and inverts the whole vagina. **Le Fort** (uterus in place) excises anterior and posterior rectangles and leaves **lateral channels** for drainage.

- Both need **high perineorrhaphy and levator myorrhaphy** and cystoscopy. If the uterus stays, evaluate the endometrium; hysterectomy removes cancer and **pyometra** risk but raises complications.
- Spare ≥**1.5 cm** of distal anterior wall, because urethral traction causes SUI. Add a sling for SUI or occult SUI.
- Success is ~90–98% and satisfaction >90%, but regret reaches **13%**.

8 · POSTERIOR COMPARTMENT

Bowel symptoms often persist after rectocele repair.

- Symptoms are bulge, splinting or digitation, trapping (**not faecal incontinence**) and dyspareunia. First exclude slow transit and **anismus**.
- **Benefit is predicted** by bulge, splinting and a large rectocele, but not by constipation. **Constipation often persists**.
- **The transvaginal route beats transanal** (recurrence 7% versus 66%).
- **Midline fascial plication**, with or without perineal body repair, is the International Consultation on Incontinence (ICI) procedure of choice. In **site-specific** repair the commonest defect is **transverse**.
- Perineorrhaphy should leave a smooth contour, because ridges cause dyspareunia.

ALGORITHM

1. Repair anterior and apical compartments first; hydrodissect and incise in the midline.
2. Dissect the rectovaginal fascia off the epithelium; if the hernia is large, place a purse-string at its base.
3. Plicate in the midline (2-0 absorbable), incorporate the perineal body, trim and close.

- **Levator plication has been abandoned** because dyspareunia reaches 50%. **Grafts** are not supported (complications RR 1.8).

9 · COMPLICATIONS ACROSS POP SURGERY — SUMMARY

Problem	Prevention / management
Ureteric injury (USLS highest, 1–11%)	Bites lateral to medial; cystoscopy after tying; remove the most lateral suture; stent; reimplant
Bladder/rectal injury	Correct plane, drained bladder, rectal finger; layered closure
Mesh exposure	Type I mesh, full-thickness flaps, supracervical hysterectomy; oestrogen or excision
Recurrence	Anterior after USLS/SSLF, posterior after SCP; most are asymptomatic
Buttock pain	Middle-third placement; observe up to 3 months; nerve injection
De novo SUI / retention	Reduction testing; selective sling; CIC; sling incision

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
CARE	RCT: SCP with or without Burch, continent women, stage 2–4	SUI 24% vs 44% at 3 months	Prophylactic Burch reduces de novo SUI
E-CARE	7-year extension	SUI 62% vs 77%; mesh exposure ~10%; composite failure 34–48%	The benefit is a delay; exposure rises with time

Trial	Design	Result	Take-home
OPUS	RCT: vaginal POP repair with MUS vs sham	27% vs 43% at 12 months; more perforation, UTI, bleeding, retention	Selective sling
OPTIMAL	RCT: USLS vs SSLF, with or without PFMT	Equal (~65%) at 2 y; PFMT no effect	Native apical repairs are equivalent
Bilateral SSLF vs abdominal SCP	RCT	Earlier failure and 3× cystocele reoperation after SSLF	Abdominal more durable
Transvaginal vs transanal rectocele	RCT	Recurrence 7% vs 66%	Repair vaginally

119 Genitourinary Complications of Mesh and Their Repair

1 · MESH, TERMINOLOGY AND REGULATION

- Unlike the abdominal wall, the pelvic floor is mobile, contaminated and close to viscera. Mesh harms are pain, exposure, extrusion and injury to nearby structures.

Amid type	Pore	Example	Behaviour
I	Macroporous >75 µm	Knitted monofilament polypropylene	**Only recommended mesh**
II	Microporous ~10 µm	Polytetrafluoroethylene (PTFE, Gore-Tex)	Encapsulated; >30% full-length slings excised
III	Macropores in microporous base	Mersilene, Mycromesh	Extrusion ~20%
IV	Submicronic	Silicone-coated	Extrusion/sinus up to 40%

KEY POINT

Why pore size matters: bacteria (~1 µm) enter pores that macrophages (~50 µm) cannot, so infection is sealed from immunity. Large pores (~1500 µm) allow fibrocollagenous integration. Woven polypropylene (Marlex) is stiff and extrudes. Mesh adds no cancer or autoimmunity.

P1 *Management of perforated mid urethral sling — IUGA/ICS terms: exposure, extrusion, perforation; "erosion" not used; trocar injury and late perforation management*

Not in the paper answer:

- IUGA/ICS **CTS (Category–Time–Site) coding**: Category 1–3 vaginal (1 no separation; 2 exposure ≤1 cm; 3 >1 cm or extrusion), 4 urinary tract, 5 bowel, 6 skin/musculoskeletal, 7 systemic; suffix A asymptomatic, B symptomatic, C infection, D abscess. Time T1 ≤48 h, T2 48 h–2 months, T3 2–12 months, T4 >12 months. Site S1 vaginal suture line, S2 vagina away from it, S3 trocar path, S4 other skin/musculoskeletal, S5 intra-abdominal.
- The FDA classed transvaginal POP mesh as **class III (2016)** and it stopped selling in **2019**; slings and sacrocolpopexy (**transabdominal POP mesh, class II**) are exempt and proven safe, whereas mini-slings are not.

2 · RISK FACTORS AND PREVENTION

- **Contraindications to vaginal mesh** are simultaneous urethrovaginal fistula repair, urethral injury or mesh perforation, and diverticulectomy. With **intraoperative urethral injury, abandon mesh** and place a sling later.

- Relative risks are prior fistula or diverticulum surgery, radiation, oestrogen-deficient tissue, smoking, obesity and age extremes; in men, radiotherapy and urethrotomy.
- POP mesh risk rises with high mesh volume, **shallow vaginal dissection**, hysterectomy, diabetes and sexual activity.

3 · EVALUATION

ALGORITHM

1. History and **prior operative notes**.
2. Speculum and pelvic exam: exposure, bands, point tenderness, levator hypertonicity.
3. If haematuria, recurrent UTI, storage LUTS or voiding pain, then cystourethroscopy.
4. If voiding dysfunction, then urodynamics (UDS).
5. Translabial ultrasound for a sling; **MRI** for sacrocolpopexy mesh (invisible on CT); CT for haematoma.

Time never excludes a complication; slings can expose decades later.

4 · MIDURETHRAL SLING COMPLICATIONS

Vaginal exposure

- It follows infection, haematoma, excess tension or superficial placement, and presents with bleeding, discharge, dyspareunia, partner pain or UTI.
- A small early exposure gets **topical oestrogen** (1 g cream nightly for 2 weeks, then 2–3 times weekly; or estradiol 10 µg tablet).
- **Excision**: raise a flap, expose the full width of mesh, pass an instrument behind, divide, excise at least **1 cm** around the defect and close. Avoid blind lateral dissection, which causes vulval haematoma after transobturator (TO) slings.

Pain

- Causes are shrinkage, tension, obturator neuralgia and pelvic floor hypertonia.
- **Immediate severe pain**: exclude haematoma and consider **early removal** before ingrowth.
- Delayed focal pain gets pelvic floor physical therapy, trigger point injection and oestrogen, then segment excision (60–90% improve). Diffuse late pain reflects central sensitisation, and excision may fail.
- Remove **as much vaginal mesh as possible** (81% pain-free). For TO **groin pain**, add a thigh incision to obturator externus.

Bladder perforation

- Cystoscopy with a **full bladder** detects trocar puncture. A missed one causes storage LUTS, haematuria, UTI, stones or vesicovaginal fistula (VVF).
- **Endoscopic** removal has low morbidity, but mesh stays in the wall and re-extrudes. **Open or robotic transvesical** removal excises 2 cm beyond the defect with two-layer closure and a catheter for 2–3 weeks; re-extrusion is minimal.

Urethral perforation

- Causes are trocar injury, over-tensioning and **urethral dilation** (avoid it; divide an obstructing sling). It presents with voiding pain, retention or haematuria.
- Repair: inverted-U flap, peel the mesh off, minimal urethrotomy, layered closure with or without a **Martius** flap, and non-overlapping suture lines. A stricture needs buccal urethroplasty.
- After excision, SUI occurs in **42%**, fewer with a simultaneous fascial sling.

Voiding dysfunction

- Midline **sling division** resolves retention in 95% and symptoms in 75%. Urgency incontinence persists in up to 76% and is de novo in up to 45%. Recurrent SUI (20–74%) is treated by bulking,

repeat MUS or fascial sling.
- Persistent symptoms need UDS, then wider excision and urethrolysis.

> **KEY POINT**
>
> **How much to remove**: retention needs midline division; exposure at least 1 cm each side; pain maximal vaginal removal, rarely arms. **Never remove asymptomatic mesh**; more removal means more bleeding and more POP and SUI recurrence.

5 · TRANSVAGINAL POP MESH

- In the largest randomised trial (n = 1358), native and mesh repair were equal at 6 years (reoperation 12% vs 10%); 8% of the mesh arm needed excision.
- Exposure is 10–12% and removal 5–8%. **Contraction** (~12%) is unique to POP mesh and causes tightness and dyspareunia. Bladder perforation is 0.7–3.5%, rectal 0.15% and fistula 0.2%.
- Treat **conservatively** (observe, oestrogen) if <1 cm, mildly symptomatic and uninfected; only **size** predicts success. After office trimming, 73% later need theatre.
- **Excision**: raise thick flaps, then clamp behind the mesh and divide, which is the **most dangerous step**. Cut arms under tension rather than chasing them, then repair natively.
- Afterwards, cystocele recurs in 19%, 20% need repeat excision and 3% develop VVF.

6 · SACROCOLPOPEXY MESH

- Exposure is ~**10.5% at 7 years**, higher with total hysterectomy (23% vs 5%) and stage 3–4.
- Features are discharge, granulation, dyspareunia, cuff abscess, bladder or bowel perforation and **discitis**; image cross-sectionally.
- Cut an **exposed suture** below the knot in the office. Oestrogen for small exposures and vaginal excision often fail; partial colpocleisis succeeds in ~50% but is unsuitable if sexually active.
- **Abdominal removal** suits failed vaginal repair, pain, infection, discitis or ureteric complication. **Remove the entire infected mesh**, including sacral sutures, because residual mesh causes sinus, abscess and fistula; bowel injury occurs in up to 29%.

7 · MALE SLINGS

- **Advance XP** causes retention in 14–21%, usually for days; incise one arm if it persists. **Virtue** causes perineal or scrotal pain in 14–17%, chronic in 0–7%.
- **Urethral extrusion** usually follows a perforation missed without intraoperative cystoscopy. Excise lateral to the urethra and repair primarily or with buccal graft, avoiding transection.

121 Urinary Tract Fistulae

1 · DEFINITION AND UNIVERSAL PRINCIPLES

- A **urinary tract fistula** is an abnormal epithelial- or mesothelial-lined connection from the urinary tract to another cavity (bowel, vessel, genital tract) or skin.
- Causes are iatrogenic (commonest in the developed world), malignancy, inflammation, infection, ischaemia, trauma and childbirth.
- It persists because **urine keeps flowing through a tract that cannot heal**, so it epithelialises. Every principle diverts flow or restores healing, and the **first repair has the best chance**.

Management (whole course) — Table 121.1	Operation (any route) — Table 121.2
Optimise nutrition	Evaluate concomitant injury; rule out malignancy
Control infection	Adequate exposure; remove all foreign material

Management (whole course) — Table 121.1	Operation (any route) — Table 121.2
Continuous drainage (catheter/stent)	Tension-free closure; vascularised edges
Eliminate obstruction	Multiple, watertight, **non-overlapping** layers
Consider malignancy — **always biopsy** if any history	Consider interposition; drain after

KEY POINT

Trap: a fistula may be the only sign of recurrent cancer, so biopsy the tract after prior malignancy or radiation.

2 · VESICOVAGINAL FISTULA (VVF) — AETIOLOGY

VVF is the commonest acquired urinary fistula (**>75%** of genitourinary fistulae).

	Developing world — obstetric (>90%)	Developed world — iatrogenic
Cause	Obstructed labour; young, small pelvis, poor nutrition	Unrecognised surgical injury; **>75%** after hysterectomy
Mechanism	Fetal head impacted for days causes pressure necrosis and a large defect	Dissection, cautery, clamp crush, bladder caught in cuff closure
Extent	Trigone, urethra, ureter often involved; "obstructed labour injury complex" (renal, rectal, orthopaedic, neurological, psychosocial)	Solitary, **supratrigonal at vaginal cuff**

- **Hysterectomy** causes VVF in ≈ 0.8/1000, lowest after vaginal and highest after laparoscopic surgery. Distorted anatomy and concomitant prolapse, incontinence or mesh surgery raise risk; immediate recognition and repair prevent fistula.
- **Universal cystoscopy at hysterectomy** raises intraoperative detection but has an inconsistent effect on delayed injury. AUGS advises it with any prolapse or anti-incontinence procedure, AAGL after laparoscopic or robotic hysterectomy, and ACOG makes no recommendation after benign hysterectomy.
- **Radiation** causes obliterative endarteritis, fibrosis and necrosis. VVF follows 0.6–2% of cervical cancer radiotherapy, typically **1.5–2 years** later (up to 30); risk is dose-dependent and worse in smokers.
- Other causes are mesh, **bevacizumab** (anti-VEGF) and cancer, and rarely pessary, intrauterine device, infection or endometriosis.

3 · CLASSIFICATION OF VVF

Of >25 systems, the WHO uses simple vs complex.

Axis	Goh	Waaldijk
Basis	Distance of distal edge from external urethral meatus	Closing (continence) mechanism and urethral involvement
Types	1: >3.5 cm · 2: 2.5–3.5 · 3: 1.5–2.5 · 4: <1.5 cm	I: closing mechanism spared · IIA: no (sub)total urethral involvement · IIB: (sub)total urethral involvement (A/B = without/with circumferential defect) · III: miscellaneous (ureteric)
Size	a <1.5 · b 1.5–3 · c >3 cm	Small <2, medium 2–3, large 4–5, extensive ≥6 cm
Tissue	i none/mild fibrosis · ii moderate/severe · iii special (radiation, ureteric, circumferential, previous repair)	—

KEY POINT

Trap: Goh type 1 is farthest from the meatus (best) and type 4 closest (worst); closer means worse closure and continence. Goh predicts closure better. Always describe size, location (trigonal vs supratrigonal), tissue loss, and ureteric and urethral involvement.

4 · DIAGNOSIS OF VVF

- The classic symptom is **continuous leakage day and night** in any position; a small tract may leak only positionally.
- **Timing points to mechanism**: a missed cystotomy leaks in the first days, thermal or crush injury at **7–10 days**, and radiation after years. Fever, flank pain or ileus suggest urinoma or ureteric injury.
- On speculum the defect is usually at the vaginal apex after hysterectomy, and vaginal fluid creatinine approximates urine creatinine. Biopsy friable tissue after cancer or radiation, and optimise oestrogen, smoking, glucose and nutrition.
- **Single dye test**: backfill ~300 mL methylene blue, insert a tampon and walk 15–60 min. Proximal stain means VVF, distal stain means urethrovaginal fistula or SUI, and dye from the cervical os means vesicouterine fistula.
- **Double dye test**: oral phenazopyridine 100–200 mg plus blue in the bladder, then a tampon. **Orange means ureterovaginal fistula (UVF)**, blue means VVF, and both means both.
- **Cystoscopy is integral** (sensitivity ~92%) and shows size, number, orifices, bladder neck, mesh and capacity, and allows biopsy. Vaginoscopy shows the vaginal side. A cystogram (lateral view) or T2 MRI helps difficult cases.
- **Upper tract imaging is mandatory**, because ureteric injury coexists in up to **12%**. Use CT urography with a delayed phase, or MR urography if contrast allergy or renal failure. **Partial ureteric obstruction with VVF means UVF until proven otherwise.** If unclear, do retrograde pyelogram and stent, or percutaneous nephrostomy (PCN).
- Urodynamics helps planning: 10–15% have SUI needing an autologous sling, and radiation may reduce capacity and compliance.

KEY POINT

Work-up order: is it urine (creatinine), where is it from (dye tests, endoscopy), and what else is damaged (hence compulsory upper tract imaging). A preoperative PCN also allows antegrade ureteroscopy to find the ureter in scar.

5 · VVF — TIMING, CONSERVATIVE CARE, REPAIR

P3 P2 P1 *VVF management options; early vs delayed repair; eponymous techniques — catheter trial (~15%, <1 cm, within 3 weeks, EAU ≥12 weeks), fulguration, sealants not recommended; timing by*

cause; vaginal vs abdominal route (~90% either); flap-splitting, Latzko, cuff excision, O'Conor, transvesical/extravesical, lap/robotic; flaps; postoperative care; diversion; outcomes

Not in the paper answer:

- Size >3 cm predicts failure, and radiation strongly predicts failure with ureteric stricture, poor compliance and sphincter deficiency, so consider diversion.
- Apical VVF (8–10 cm from introitus) lies beyond Martius reach, so use peritoneum or omentum.

Setting	Timing of repair	Rationale
Obstetric (traditional)	≥3 months, catheter meanwhile	Inflammation settles, necrosis sloughs
Iatrogenic, clean tissue	Early; European Association of Urology (EAU): no minimum delay if no infection or recent radiation	Comparable closure; less distress

Setting	Timing of repair	Rationale
Radiation-induced	≥**12 months** after radiation ends	Late injury still evolving

- **Catheter** drainage is favourable for tracts <1 cm catheterised within **3 weeks**, and unfavourable at ≥6 weeks or with radiation or malignancy. The EAU trial lasts ≥**12 weeks**. Sealants are possible for <1 cm in poor surgical candidates.
- **Location**: within **1 cm** of a ureter, stent it; if the ureter is involved, reimplant abdominally. Bladder neck involvement needs reconstruction, plus a pubovaginal sling if needed.
- **Route**: there are no RCTs, and first-attempt closure is ≈ **90%** either way. Transvaginal repair has less blood loss, pain and stay.
- Abdominal repair suits large fistulae (≥4 cm primary, ≥2 cm recurrent), urethral, bladder-neck or ureteric involvement, a non-capacious vagina, radiation, or no lithotomy. Vaginal repair risks shortening and dyspareunia, whereas abdominal repair spares sexual function.

ALGORITHM

1. **Transvaginal flap-splitting (Table 121.5)**
2. Cystoscopy: stent close ureters; wire through the tract.
3. Suprapubic catheter (SPC) and urethral catheter; episiotomy if narrow.
4. Foley ≥8 Fr through the fistula for traction; hydrodissect.
5. **J- or U-incision** with a 1–3 mm cuff; raise wide anterior and posterior flaps.
6. Layer 1: 3-0/4-0 delayed absorbable. Layer 2: **perpendicular** to layer 1.
7. Leak test 200–300 mL; Martius or peritoneal flap if needed.
8. Advance the opposite flap ≥2 cm beyond (layer 3); cystoscopy for efflux.

- **Postoperative drainage is the most important part**: urethral catheter, often with SPC, for ≥10–14 days. Give an anticholinergic or β3-agonist for spasms, remove stents at 4–6 weeks, and advise pelvic rest for ~6 weeks.
- **Latzko** leaves the bladder unopened and the tract unexcised: denude 1–2 cm and close in two or more layers. It succeeds in ≥90% of non-irradiated cases and suits close orifices, but shortens the vagina.

ALGORITHM

1. **O'Conor (Table 121.7)**
2. Ureteric catheters; low midline extraperitoneal approach.
3. Sagittal dome cystotomy; **bivalve** down around the fistula; excise the tract.
4. Separate bladder from vagina ≥2–3 cm beyond; close the vagina.
5. Place **omentum or peritoneum** beyond the vaginal suture line.
6. Close bladder in 1–2 layers; SPC with or without urethral catheter; fill test.
7. Drain 10–14 days, then cystogram.

- O'Conor causes more de novo urgency and LUTS from long suture lines. Transvesical repair suits low-lying fistulae, and extravesical repair suits supratrigonal fistulae. Robotic O'Conor uses omentum or peritoneum, with indocyanine green to check edge vascularity.

6 · INTERPOSITION FLAPS

- Flaps reinforce, fill dead space and import blood supply. Use them for **complex VVF** (radiation, recurrent, ischaemic obstetric, poor tissue); an RCT showed **no benefit of Martius** in simple VVF.
- **Martius** is supplied by the internal pudendal (inferior), external pudendal (superior) and obturator (lateral) vessels. Keep the **superior** pedicle for VVF (inferior for rectovaginal). It reaches mid-distal VVF, bladder neck and urethra, but **not the apex**.
- A vaginal peritoneal flap suits the apex, and **omentum** suits radiation, recurrent or large fistulae. Rectus, gracilis and Singapore flaps are a last resort.

- After repair, residual incontinence (predicted by Goh 3–4 and more than one repair) drives quality of life, and ICS advises physiotherapy for all obstetric fistula. Divert if the fistula is unrepairable.

7 · URETEROVAGINAL FISTULA (UVF)

P3 *Urological complications in OBS-GYN and uretero-vaginal fistula — distal ureteric injury at hysterectomy, danger sites, normal voiding + leak within 4 weeks, double dye/CTU/RGP, stent within 2*

weeks for 4–6 weeks, reimplant ± psoas hitch/Boari, ≥90%, follow 30 months

Not in the paper answer:

- Up to **25%** of VVF have a coexisting UVF. Prophylactic stents are selective (e.g. caesarean hysterectomy for placenta accreta).

ALGORITHM

1. Suspect UVF; exclude VVF; confirm with CTU or retrograde pyelogram.
2. Stent; remove at 4–6 weeks and reimage.
3. If resolved, stop. If persistent or stenting fails, then do ureteroneocystostomy.

KEY POINT

UVF: wet but voids normally. The ureter heals over a stent if reached early, so the detection-to-stent interval largely decides whether surgery is needed.

8 · VESICOUTERINE FISTULA (VUF)

- It follows **low transverse caesarean** and joins the uterine body above the internal os to the posterior bladder; a competent cervix **preserves continence**.
- **Youssef syndrome** is menouria, amenorrhoea, first-trimester miscarriage and continence. Bladder endometriosis is the differential of menouria.
- Dye appears from the cervical os, and cystoscopy shows a posterior supratrigonal defect; add cystogram, hysterosalpingography or MRI.
- Spontaneous closure is ~5%. After 3–4 months, separate transabdominally and close in layers with omentum or peritoneum, or do hysterectomy if the family is complete. Deliver later pregnancies by caesarean.

9 · URETHROVAGINAL FISTULA

P3 *Urological complications of OBGY surgery and urethrovaginal fistula — diverticulectomy commonest Western cause (20–25%), distal vs proximal symptoms, flexible urethroscopy, ~20% separate VVF, flap*

repair + Martius, autologous sling never synthetic, SPC ≥14 days, closure 90–95% vs continence 50–80%

10 · RECTOURETHRAL FISTULA (RUF)

P3 *Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management — post-prostate-*

cancer treatment, faecaluria/pneumaturia, biopsy, colostomy first, transperineal gracilis vs York-Mason vs abdominal, salvage diversion

Martini	Category
Stage (size)	I ≤1.5 cm · II >1.5 cm · III any size with sphincter damage
Grade (tissue)	0 surgical injury only · 1 prior radiation/energy · 2 salvage radical prostatectomy (RP) or salvage ablation

11 · VESICOENTERIC FISTULA (VEF)

- Colovesical is commonest. M : F is **3 : 1** because the uterus shields the bladder.
- Causes in order are **diverticulitis**, **Crohn disease** (ileovesical) and colorectal cancer (10–15% of colovesical fistulae), then bladder or prostate cancer, radiation and appendicitis.
- It causes **pneumaturia** (50–70%), faecaluria and polymicrobial UTI. **Gouverneur syndrome** is suprapubic pain, frequency, dysuria and tenesmus.

Test	Role
Poppy seed test	Highest detection of presence; cheap; no location
CT (preferred)	Air in bladder, wall thickening, adherent bowel; shows cause
Cystoscopy	Nonspecific erythema/oedema
Colonoscopy	Poor for tract; **excludes cancer**
MRI	Crohn, anatomy

- Treat conservatively if symptoms are minimal and disease benign (especially Crohn); spontaneous closure is <2%.
- **Resect the diseased bowel with the fistula** with single-stage anastomosis; use Hartmann for abscess, contamination, advanced cancer or radiation. Leave or oversew a small bladder defect, do **partial cystectomy** if malignant, and interpose omentum.

> **KEY POINT**
> VEF is a bowel disease presenting in the bladder: remove the diseased bowel, and the bladder usually heals itself.

Renocolic fistula: spontaneous ones are mostly renal (xanthogranulomatous pyelonephritis [XGP]) and need nephrectomy; small iatrogenic ones after PCNL settle with antibiotics, drainage and nil by mouth.

12 · UROCUTANEOUS FISTULAE

- **Vesicocutaneous** fistula follows long-term SPC, open prostatectomy or radiation, and persists with infection, foreign body, tumour or bladder outlet obstruction (**BOO**). Drain and relieve BOO; if unhealed, do fistulectomy or partial cystectomy.
- **Urethrocutaneous** fistula is the commonest hypospadias complication (up to 20%). Wait ≥**6 months** and exclude distal stricture. Close <1 cm locally; ≥1 cm or redo needs vascularised flaps (**tunica vaginalis** better than dartos).
- **Nephrocutaneous** fistula follows stones, XGP, prior surgery or tuberculosis. The kidney is non-functioning in ~90%, so do **nephrectomy**.

13 · PUBOVESICAL FISTULA (PVF)

- A tract from bladder, prostate or urethra to the pubic symphysis causes **osteomyelitis**. It follows prostate radiotherapy plus repeated outlet procedures, median ~10 years after radiation, and also artificial urinary sphincter surgery.
- It presents with **pubic pain on walking**, abscess, urosepsis or sinus. **MRI is most sensitive and specific**; bone biopsy guides antibiotics because urine culture is unreliable.
- Conservative care usually fails. Debride bone, excise the fistula, fill with a rectus or omental flap, and give 4–6 weeks of antibiotics; if bladder or sphincter function is poor, do **cystoprostatectomy and diversion**.

14 · UROVASCULAR FISTULAE

- **Arteriocalyceal** fistula follows PCNL, transplant or trauma; treat by **angioembolisation**.

- **Ureteroarterial fistula (UAF)** is an emergency. Most (85%) are secondary to long-term stents, pelvic surgery, radiation, cancer or vascular grafts. It presents with **intermittent haematuria**, and ~20% are unstable.
- Diagnose with **angiography**, provoking by stent removal only with endovascular treatment ready; a negative CT angiogram does not exclude it.
- Treat with an **endovascular stent graft** with or without internal iliac embolisation; use open surgery for infection, abscess or enteric involvement.

15 · RENAL URINE LEAK AND ANASTOMOTIC LEAK

- After **trauma**, a leak occurs in 1–7%. Use CT with a 10-min delayed phase; a non-opacified distal ureter suggests ureteropelvic junction avulsion. Most resolve, and persistent leaks need stent or nephrostomy.
- After **partial nephrectomy** (0.5–20%), a central tumour predicts leak. Drain, adding stent or nephrostomy if needed.
- **Vesicourethral anastomotic leak after RP** (Patil): I extraperitoneal <6 cm, II extraperitoneal ≥6 cm, III intraperitoneal. Prolong the catheter, then withdraw the drain, then place stents or nephrostomies; reoperate last.

16 · FISTULA AFTER GENDER-AFFIRMING SURGERY

- After phalloplasty or metoidioplasty, urethrocutaneous fistula and stricture are common; **primary colpectomy** roughly halves fistula. Small ones close with a catheter. After transfeminine surgery, neovaginal fistula is rare and is repaired with interposition after diversion.

17 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Martius flap in simple obstetric VVF	RCT, repair ± Martius	No difference in closure; ~⅓ donor-site pain/numbness	Reserve Martius for complex fistula
Prophylactic ureteric stents in gynaecological surgery	RCT (+ meta-analysis)	No/small reduction in injury (number needed to treat 224)	Selective, not routine, stenting

122 Bladder and Female Urethral Diverticula

1 · BLADDER DIVERTICULA — DEFINITION AND PATHOLOGY

P2 *Bladder diverticulum. — definition, congenital vs acquired, investigations, management, diverticulectomy steps, T3a staging*

Not in the paper answer:

- Diverticular cancer: radical cystectomy gives the same stage-for-stage survival; upstaging is common.
- A diverticulum is urothelium herniating through the muscularis propria; an outer pseudocapsule gives the surgical plane.
- With **no functional muscle, the sac cannot contract**, so post-void residual (PVR) is high and stasis leads to infection, stones and malignancy.
- About 90% occur in adults; M : F ≈ **9 : 1**.

Feature	Congenital	Acquired
Age / number	<10 years; solitary	Men >60; often multiple
Site	Posterolateral to ureteric orifice (~90% near ureterovesical junction [UVJ]); vesicoureteral reflux (VUR) common	Ureterovesical hiatus commonest; anywhere

Feature	Congenital	Acquired
Cause	Detrusor weakness at UVJ	Bladder outlet obstruction (**BOO**) or neurogenic — almost always
Wall / presentation	Smooth; acute urinary tract infection (UTI)	Trabeculated; incidental (lower urinary tract symptoms [LUTS], haematuria, UTI)
Malignancy	Not increased	Increased

- Up to 60% of congenital cases have a syndrome, neuropathic bladder or BOO, e.g. **Menkes**, **Williams**, **Ehlers-Danlos**, prune-belly or posterior urethral valves.
- Acquired cases are due to benign prostatic hyperplasia in ~**70%**; in women, causes include dysfunctional voiding, prolapse, bladder neck obstruction and post-sling obstruction.
- A **Hutch diverticulum** is paraureteric, in a neuropathic bladder with VUR.

KEY POINT

Congenital: weak wall, normal pressure. Acquired: normal wall, high pressure, so an acquired diverticulum sends the search back to the outlet.

2 · BLADDER DIVERTICULA — EVALUATION

- Symptoms are nonspecific (incomplete emptying, double voiding).
- Do rectal examination, PSA, urinalysis, culture, **cytology**, endoscopy and upper and lower tract imaging, and seek BOO or occult neuropathy.
- Voiding cystourethrogram (**VCUG**) shows size, VUR and emptying. **Paradoxic enlargement** means the sac fills on voiding and refills the bladder afterwards.
- CT or MRI shows number, masses and relation to ureters and rectum.
- Hydroureteronephrosis (HUN) occurs in ~7% of adults, from compression, underlying dysfunction or VUR; the pelvic ureter deviates **medially**.
- Video-urodynamics (**VUDS**) shows a **pressure-sink effect**: contractility looks falsely low, and a raised PVR may be diverticular urine.
- **Cystoscopy** inspects the whole sac with cytology from it. Biopsy with extreme care, as perforation seeds tumour.
- With VUR and a congenital diverticulum, reimplant **selectively**. Treating the cause may make excision unnecessary.

KEY POINT

Two jobs: find the cause (outlet or neurogenic) and exclude cancer in a sac each test sees imperfectly, so tests are combined.

3 · MALIGNANCY IN BLADDER DIVERTICULA

- Prevalence is 0.8–10%, driven by stasis and inflammation. Urothelial carcinoma is **70–80%** and squamous cell carcinoma (SCC) 20–25%.
- With **no muscularis propria**, spread is early and extravesical, transurethral resection (TUR) risks perforation, and staging is inaccurate.
- **Radical cystectomy** gives the same stage-for-stage survival. Partial cystectomy or diverticulectomy suits selected low-grade, low-stage tumours.

KEY POINT

Treat a diverticular tumour as potentially invasive from the outset; missing muscle explains early spread, perforation risk and inaccurate staging.

4 · BLADDER DIVERTICULA — MANAGEMENT

Treat the outlet first; operate for complications, not size.

ALGORITHM

1. Assess with history, cystoscopy, cytology and VCUG.
2. If a lesion is suspicious, then biopsy and give cancer therapy.
3. If not, image the upper tract. If HUN or symptoms, then do urodynamics and treat BOO, then reassess.
4. If a problem from the diverticulum persists: fit → diverticulectomy; unfit → clean intermittent catheterisation (CIC).
5. If asymptomatic and uncomplicated, then survey.

- **Diverticulectomy indications**: refractory LUTS, relapsing UTI, **stones**, **carcinoma or premalignancy**, upper tract deterioration (obstruction, VUR). **Size alone is not an indication.**
- **Treat the outlet first or simultaneously**; a small sac may then empty, and concomitant prostatectomy is common.
- If observing, counsel about cancer and survey with symptoms, cytology and cystoscopy. The unfit with poor emptying need CIC or a catheter.
- **Endoscopic** treatment for the unfit resects the neck to muscle, but large sacs do poorly.
- Diverticulectomy is open, laparoscopic or robotic: use an ipsilateral ureteric catheter, dissect in the pseudocapsule, close the neck in layers, and reimplant a ureter entering the sac.

5 · FEMALE URETHRAL DIVERTICULUM (UD) — ANATOMY AND AETIOLOGY

P1 *Female urethral diverticula — Routh theory, posterolateral ostium, three Ds ~5%, stones 4–10%, adenocarcinoma, imaging sensitivities (MRI 97%), L/N/S/C3, diverticulectomy principles, autologous sling only, recurrence 10–22%, de novo SUI 7–16%*

Not in the paper answer:

- Configuration is simple in ~73%, saddlebag in ~17% and circumferential in ~10%; ~50% are mid-urethral and 6–19% multiple.
- Carcinoma in UD is **adenocarcinoma** > urothelial > SCC. Local excision alone recurs, so use anterior exenteration or cystourethrectomy with or without radiotherapy.
- **Periurethral glands** lie posterolaterally, with ducts draining into the **distal third**; UD lie between the leaves of the urethropelvic ligament. The striated sphincter is mid-urethral, so plan repair for continence.
- In the gland theory, a gland abscess herniates into the fascia, usually **ventrally**, then ruptures into the lumen; about a third are proximal, however.

6 · UD — PRESENTATION AND EVALUATION

- All three Ds occur in only ~**5%**. Features are irritative LUTS, pain, recurrent UTI, mass (~⅓), incontinence (~⅓), haematuria and discharge; 10–20% are asymptomatic. **Size does not track symptoms**, and UD is often misdiagnosed for years.
- In pregnancy, give antibiotics or aspirate, and repair after delivery.
- A mass lies 1–3 cm inside the introitus, and a **hard mass suggests stone or cancer**. Check atrophy, stress test and prolapse.
- **Cystourethroscopy** shows the ostium at **4 and 8 o'clock** mid-urethra in 15–89%; not seeing it must not stop work-up.
- **VUDS** shows SUI in up to 50% and separates true SUI from dribbling from the sac.
- **MRI is best** (T2 high, maps extent) but may miss small cancers. Ultrasound shows an anechoic sac, and VCUG underestimates size.

KEY POINT

UD symptoms mimic common LUTS and do not track size, so suspicion and examination make the diagnosis. Imaging confirms and **maps** it, especially extension towards the bladder neck (bladder or ureteric injury, postoperative incontinence).

7 · DIFFERENTIAL DIAGNOSIS OF PERIURETHRAL MASSES

Lesion	Clue	Treatment
UD	Mid/proximal; communicates	Excision, reconstruction
Leiomyoma	Firm, mobile, solid; oestrogen-dependent	Transvaginal enucleation
Skene cyst/abscess	Lateral to and distorts meatus; non-communicating	Aspiration, marsupialisation, excision
Gartner duct cyst	Anterolateral; may receive ectopic ureter — image upper tract	Observe, sclerotherapy, excision
Vaginal wall cyst	Müllerian commonest	Excise if symptomatic
Urethral prolapse	Red "doughnut"; prepubertal, postmenopausal	Oestrogen; excision
Caruncle	Red, posterior meatal lip; postmenopausal	Oestrogen; excise if atypical
Bulking agent	Injection history	Excise if symptomatic

Skene gland adenocarcinoma may raise PSA.

KEY POINT

Two questions sort most periurethral masses: does it communicate (only UD), and where is it relative to the meatus (Skene, prolapse and caruncle lie at the meatus)? A firm, mobile, solid mass is a leiomyoma.

8 · CLASSIFICATION AND SUI

System	Basis	Use
L/N/S/C3 (Leach)	Location, Number, Size, Configuration, Communication, Continence	Standard; not validated
Ginsburg–Genadry	Location	Distal: marsupialisation; proximal: excision
Leng–McGuire	Periurethral fascia present?	Absent = **pseudodiverticulum**, needing flap/graft

- Significant de novo stress urinary incontinence (SUI) occurs in **7–16%**; risks are size ≥30 mm, a proximal or circumferential sac and a short urethra. Most settles within 12 months.
- About 50% of preoperative SUI improves after excision, so operate only for **bothersome** SUI, using an **autologous pubovaginal sling, never synthetic**.

9 · SURGICAL REPAIR OF UD

- Operate for symptoms, stones or suspected cancer; treat the asymptomatic or unfit with suppressive antibiotics and postvoid stripping.
- **Spence-Duckett marsupialisation** suits only distal-third UD in women who are not sexually active.
- Principles (Box 122.2): vascularised flap, separate fascia, excise ostium and entire sac, watertight multilayer non-overlapping closure, close dead space, and **preserve or create continence**.

ALGORITHM

1. Lithotomy and Foley; suprapubic catheter (**SPC) at the start**, never the end (it would cross the repair).
2. Make an **inverted-U incision** and raise the vaginal flap off the fascia.
3. Incise the fascia **transversely** and raise fascial flaps.
4. Excise the entire sac to the ostium, including the inflamed urethral edge.
5. Close the urethra watertight over a catheter (as small as 12 Fr).
6. Close fascia **perpendicular** to it, obliterating dead space.
7. If tissue is poor or fascia attenuated, then add a **Martius** flap.
8. Advance the vaginal flap (3 layers; 4 with Martius); antibiotic pack.

- A pseudodiverticulum needs a Martius flap or fascial graft. Do a **pericatheter VCUG at 14–21 days**, repeated weekly if leaking.

10 · COMPLICATIONS

Complication	Incidence	Notes
Recurrent UD	**10–22%** (pooled ~10%)	Incomplete excision, dead space; circumferential, complex, proximal, multiple
Incontinence	1.7–16%	Conservative 12 months, then autologous sling
Urethrovaginal fistula	0.9–8.3%	Distal: may need no repair; proximal: repair ± Martius
Stricture	0–5%	—
Recurrent UTI	up to 31%	Investigate for recurrence
Others	—	Distal urethral necrosis, bladder/ureteric injury, dyspareunia

Redo excision gives less relief and more fistula, so image persistent symptoms for recurrence or stricture.

KEY POINT

Complete sac removal, watertight closure, off-axis fascia and obliterated dead space prevent the three failures: recurrence, fistula, incontinence.

Feature	Bladder diverticulum	Female UD
Origin	Herniation through detrusor	Infected periurethral gland ruptured into lumen
Cause to treat	BOO / neurogenic	Infection; coexisting SUI
Commonest cancer	Urothelial (then SCC)	Adenocarcinoma
Key study	Cystoscopy + cytology; VCUG	MRI
Operate for	Complications, not size	Symptoms or complications
Operation	Diverticulectomy + outlet treatment	Transvaginal excision, layered reconstruction

77 Sexual Function and Dysfunction in the Female

P3 *Female sexual dysfunction — definition, prevalence, response models, DSM-5 vs ICSM/ISSWSH, causes, PLISSIT/FSFI, treatment overview by domain*

Not in the paper answer:

- Flibanserin: about half respond by 8 weeks; REMS. Bremelanotide: ≤8 doses/month; hyperpigmentation. Testosterone: stop at 6 months if no benefit.
- Genital-sparing cystectomy allows sex at about 6 vs 14 weeks; combined anterior and posterior repair increases dyspareunia; TOT spares the dorsal clitoral nerve.

1 · FUNCTIONAL ANATOMY

The **distal vagina, clitoris and urethra** share blood supply and nerves and respond as one unit, the **clitoral complex**, so urethral or vaginal surgery can change sexual function.

- The **clitoris** is the penile homologue, **separate from the urethra**; its two corpora continue as crura on the ischiopubic rami.
- The glans is very sensitive but **not erectile**. The dorsal nerves lie **superficially on the dorsum**, outside the tunica, so hood or clitoral surgery can injure them.
- **Erectile** tissue is the clitoral body, crura and vestibular bulbs; labia minora, glans, periurethral tissue and vaginal walls are vascular but do not engorge.
- **Skene** glands are the prostate homologue (PSA positive) and **Bartholin** glands the bulbourethral homologue. MCQ The G-spot has no agreed anatomical basis.
- Nerve supply of the labium majus and the glans draining **directly to deep inguinal nodes** are minor exam points.

2 · PHYSIOLOGY OF THE SEXUAL RESPONSE

ALGORITHM

1. Touch on the clitoral complex travels in pudendal (somatic) afferents to **S2–S4**.
2. Parasympathetic efferents (nervi erigentes, then cavernous nerves) dilate arterioles, causing engorgement and lubrication.

The reflex has **somatic afferents and visceral efferents**. **Orgasm** is cerebral, with **sympathetic** output (T12–L1) contracting the uterus and pudendal reflexes contracting pelvic striated muscle.

> **KEY POINT**
>
> Pudendal injury removes the **trigger**; pelvic plexus or cavernous nerve injury removes the **response**.

ALGORITHM

1. Neuronal nitric oxide synthase (nNOS) makes nitric oxide (NO), which activates guanylate cyclase to make cyclic GMP (cGMP).
2. cGMP lowers Ca^{2+}, so smooth muscle relaxes and tissue engorges.
3. Phosphodiesterase type 5 (PDE5) degrades cGMP; vasoactive intestinal polypeptide (VIP) works with NO.

Desire balances **excitation** (dopamine, noradrenaline, melanocortins, oxytocin, testosterone, oestrogen) against **inhibition** (serotonin, opioids, prolactin, progesterone); licensed HSDD drugs act on this balance.

	Oestrogen	**Testosterone**
Role	Maintains the **substrate**: epithelium, collagen, lubrication, acidic pH, NOS/VIP	Drives **desire**; inactive when bound to sex hormone–binding globulin (SHBG)
Deficiency	Thin, dry, fibrotic vagina causing dyspareunia; risk rises at oestradiol **≤50 pg/mL**	Lower libido, arousal, sensation and orgasm
Treatment risk	—	Virilisation: early reversible (acne, hirsutism); late irreversible (voice, alopecia, clitoromegaly)

Model	**Stages**	**Key idea**
Masters & Johnson (1966), linear	Excitement, plateau, orgasm, resolution	**No desire phase**
Kaplan (1977), linear	Desire, arousal, orgasm	Adds desire

Model	Stages	Key idea
Whipple & Brash-McGreer, circular	Seduction, sensations, surrender, reflection	Reflection feeds next seduction
Basson (2002), circular	Neutrality, receptivity, arousal, **responsive desire**, satisfaction	Spontaneous desire **not needed**; intimacy is the motive

> **KEY POINT**
> Under Basson, lack of spontaneous desire (common after menopause) is not a dysfunction, which is why DSM-5 merged desire and arousal.

3 · CLASSIFICATION

DSM-5 requires **≥6 months** and **clinically significant distress**; low desire without distress is not a disorder.

Domain	DSM-IV-TR	DSM-5
Desire and arousal	HSDD; female sexual arousal disorder (FSAD); sexual aversion disorder	**Female sexual interest/arousal disorder (FSIAD)**, **≥3 of 6**: less interest; fewer fantasies; less initiation/receptivity; less pleasure, response to cues or sensation (in 75–100% of encounters). Aversion dropped
Orgasm	Female orgasmic disorder (FOD) after normal excitement	FOD: delay, infrequency, absence **or lower intensity** in **75–100%**; normal-excitement requirement removed
Pain	Dyspareunia; vaginismus	**Genito-pelvic pain/penetration disorder (GPPPD)**, ≥1 of: penetration difficulty; pain; fear of pain; pelvic floor muscle (PFM) tensing

- **ICSM 2016** keeps HSDD and FSAD **separate** and adds **persistent genital arousal disorder (PGAD)** and orgasmic illness syndrome; **ISSWSH** adds a urological **genital arousal disorder**.
- About **40%** have a lifetime sexual problem but only about **8–12%** a distressing disorder. **HSDD is commonest**, FOD second.

4 · CAUSES (BIOPSYCHOSOCIAL)

- **Biological**: diabetes, hypertension, coronary disease, thyroid disease, hyperprolactinaemia, renal failure, spinal cord injury (SCI), multiple sclerosis (MS), vulval dermatoses, incontinence, pregnancy, breastfeeding and menopause.
- **Psychological**: depression (bidirectional, so screen every HSDD), anxiety, body image, abuse, conflict and the **partner's erectile dysfunction or premature ejaculation**.
- **Drugs**: SSRIs, SNRIs, antipsychotics, antihypertensives, antiandrogens, GnRH agonists, opiates and alcohol. MCQ Combined hormonal contraceptives and oral oestrogens raise SHBG.
- **Genitourinary syndrome of menopause (GSM)** causes dryness, burning, dyspareunia (often **vestibular**), urgency, dysuria and recurrent UTI. Dryness causes pain, which lowers desire, so **treat the tissue first**.
- **Cancer treatment** harms by castration, anti-oestrogens (aromatase inhibitors cause severe GSM), pelvic radiotherapy (stenosis) and nerve injury.

5 · EVALUATION

Women rarely volunteer sexual problems, so the **clinician must ask**, gender-neutrally.

ALGORITHM

1. **P**ermission.
2. **L**imited **I**nformation: cycle, ageing, drugs.
3. **S**pecific **S**uggestions: lubricants, **vaginal oestrogen**, positions, vibrators.
4. **I**ntensive **T**herapy means **referral** (PLISSIT).

- **DSDS** (Decreased Sexual Desire Screener) is a **validated** 5-item screen; yes to items 1–4 with no contributing cause means **generalised acquired HSDD**. The ACOG BSSC-W is not validated.
- **FSFI** (19 items, 6 domains) is most used; FSDS-R measures **distress**; **PISQ-12/IR** is the **only validated** tool for prolapse and incontinence.
- History: lifelong or acquired, generalised or situational, the **order problems appeared**, drugs, surgery, LUTS, prolapse and the **partner's** function.
- Examination is essential with pain. **pH ≥4.5** without infection or semen means atrophy. Do a **Q-tip test** of the vestibule and **retract the hood** for adhesions.
- Lichen sclerosus leads to introital stenosis; lichen planus (**Wickham striae**) to vaginal stenosis.
- **Overactive** PFM causes dyspareunia and **underactive** PFM causes incontinence and prolapse; fornix tenderness suggests endometriosis.
- Labs are rarely needed. Do **not** measure a **random testosterone**: there is no female range and no level predicts HSDD.

6 · HYPOACTIVE SEXUAL DESIRE DISORDER

- ISSWSH defines HSDD as ≥6 months of lost desire **with distress**, not due to drug, illness, pain, abuse or partner. **Exclude relationship conflict first.**
- HSDD peaks in **midlife** and is defined by distress, not hormone level.

ALGORITHM

1. Screen (DSDS), then history and examination.
2. **Correct modifiable factors**: drugs, depression, GSM, pain, partner's ED.
3. Education; sex or couples therapy.
4. If **generalised acquired** HSDD persists, then drug; review at 8 weeks.
5. **Refer** if situational HSDD fails counselling, or psychiatric illness or sexual trauma.

	Flibanserin	**Bremelanotide**
Mechanism	Central **5-HT1A agonist / 5-HT2A antagonist**: lowers serotonin, raises dopamine/noradrenaline (removes the brake)	Central **melanocortin agonist** releasing dopamine (presses the accelerator)
Licence	FDA: only **premenopausal** acquired generalised HSDD	
Dose	**100 mg at bedtime** daily; about half respond by 8 weeks	**1.75 mg subcutaneous** ≥45 min before sex; ≤1/24 h, **≤8/month**
Side effects	Dizziness, somnolence, nausea	Nausea, flushing, headache, **hyperpigmentation**
Cautions	**Alcohol** (hypotension, syncope); strong CYP3A4 inhibitors; REMS restricted prescribing	Uncontrolled hypertension or cardiovascular disease

- **Transdermal testosterone** (off-label) improves desire, arousal, orgasm and satisfaction in **postmenopausal HSDD**, with no cardiovascular signal but unknown long-term safety.
- The 2019 Global Consensus targets **premenopausal physiological levels**, checks levels at baseline and 3–6 weeks, and says **stop at 6 months** if there is no benefit. Bupropion (off-label) is an alternative.

7 · AROUSAL DISORDER AND PGAD

- **Subjective and genital arousal correlate poorly**, so vasodilators raise flow without felt arousal. Use mindfulness-based sex-focused CBT.
- **Transdermal testosterone** has the best evidence. PDE5 inhibitors help only with neurovascular impairment (type 1 diabetes, SCI, MS, SSRI-induced). **Bupropion 150 mg twice daily** treats SSRI-induced dysfunction.
- **PGAD** is distressing genital arousal for hours to days, **not relieved by orgasm** and without desire. Causes include Tarlov cysts, pelvic varices and **stopping an SSRI**; treat with CBT, duloxetine or pregabalin.

8 · FEMALE ORGASMIC DISORDER

- FOD is a **diagnosis of exclusion requiring distress**: exclude drugs, disease, stress and **inadequate stimulation** (including the partner's ED/PE).
- **No orgasm at coitus without clitoral stimulation is normal variation.**
- **Education is the cornerstone**, with directed masturbation, sensate focus and mindfulness. If arousal is intact, increase stimulation (vibrators; MCQ EROS clitoral vacuum is the only FDA-cleared FSD device).

ALGORITHM

1. **SSRI-induced dysfunction**: was it present before treatment?
2. If not, wait for tolerance (~4 months).
3. If it persists at 3–6 months, then lower the dose, or switch to or add **bupropion** or mirtazapine; consider sildenafil.

9 · GENITO-PELVIC PAIN / PENETRATION DISORDER

- **Dyspareunia** is superficial or deep (endometriosis, prolapse, pelvic floor). **Vaginismus** is involuntary contraction preventing penetration.
- **Vulvodynia** is vulval burning pain for **≥3 months** without a cause; it is localised or generalised, provoked or unprovoked.
- **Provoked vestibulodynia (PVD)** is commonest (about **10–12%** of women). Primary PVD starts at first penetration; secondary PVD (about two-thirds) follows pain-free sex.
- The mechanism is vestibular nerve proliferation with a **hypertonic pelvic floor** and fear-avoidance.
- **Clitorodynia**: hood adhesions trap keratin and form painful pseudocysts, so **examine the glans routinely**.

ALGORITHM

1. Education and vulval care.
2. Lidocaine 5% or EMLA **10 min before intercourse**.
3. **Multimodal pelvic floor physical therapy** (myofascial release, down-training, biofeedback, graded **dilators**) with psychological/couples therapy.
4. If hypo-oestrogenic, then vestibular oestrogen.
5. If localised PVD is refractory, then **vestibulectomy**.

- **Multimodal physical therapy is standard of care** for PVD; EMG biofeedback is best if the pelvic floor is overactive.
- Oestrogen deficiency **increases** vulval nerve density, so **topical beats systemic and vestibular beats vaginal**.
- Second-line options are amitriptyline, gabapentin and intralevator botulinum toxin. **Vestibulectomy** beats non-surgical therapy in localised neuroproliferative disease.

Treatment of GSM

ALGORITHM

1. Non-hormonal lubricants/moisturisers and regular activity (NAMS first line).
2. Then **low-dose vaginal oestrogen** (minimal absorption, no progestogen) or vaginal DHEA (prasterone 6.5 mg nightly).
3. If vaginal products are impossible, then ospemifene 60 mg orally (SERM, vaginal agonist; VTE risk).
4. If vasomotor symptoms, then systemic hormone therapy.

KEY POINT

After **breast cancer**, use non-hormonal measures first; vaginal oestrogen only after oncology discussion (aromatase inhibitors).

10 · PELVIC SURGERY AND SEXUAL FUNCTION

Surgery harms by three routes: **nerves** (lateral vaginal wall, pelvic plexus), **clitoral blood supply** (distal urethra) and **vaginal shape**. Counsel **before and after**.

- **Radical cystectomy**: loss of desire is commonest, and about half can have intercourse. Preserve function by saving the bundles on the **lateral vaginal walls**, dividing the urethra at the **bladder neck**, and keeping vaginal dimensions.
- **Genital-sparing** cystectomy (if oncologically safe) lets sex resume at **~6 vs 14 weeks**; diversion type is irrelevant. **Never trade oncological control.** See Ch 101
- **Hysterectomy** for benign disease leaves function unchanged or better; after **rectal surgery** dysfunction is common, and nearly universal after abdominoperineal resection.
- **Prolapse repair**: bother and body image, not stage, predict function. Dyspareunia is commoner after vaginal than abdominal repair, and after anterior mesh than **sacrocolpopexy**.
- **Combined anterior and posterior repair increases dyspareunia**, so stage it in sexually active women. **Keep length, avoid excess skin excision, preserve the introitus.** See Ch 119
- **SUI surgery**: about a third improve and most are unchanged; **curing coital incontinence** predicts improvement. Retropubic tape (**TVT**) may denervate the vaginal wall, while transobturator tape (**TOT**) spares the dorsal clitoral nerve. Dyspareunia is a strong reason for sling removal.
- **Coital incontinence** (about 1 in 10 incontinent women): leakage at penetration is stress; at orgasm, detrusor overactivity.
- **Bladder pain syndrome causes the worst FSD**, through pain, PFM overactivity and vulvodynia. **Sacral neuromodulation** improves sexual function whatever the cause.
- **Genital cosmetic surgery** has poor evidence; screen for **body dysmorphic disorder**.

11 · SPECIAL POPULATIONS

Ask about orientation and gender identity; offer fertility preservation before transition. Oestrogen risks thromboembolism; testosterone risks erythrocytosis (haematocrit **>50%**), so check haematocrit and testosterone 3-monthly in year 1, then 1–2 times a year. SCI impairs sensation, lubrication and orgasm. See Ch 126

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Flibanserin phase III	RCTs vs placebo, premenopausal	More satisfying events and desire, less distress	First FDA-approved drug for HSDD
RECONNECT (bremelanotide)	2 phase III RCTs	More desire, less distress	On-demand option for premenopausal HSDD

Trial	Design	Result	Take-home
Bupropion for SSRI-induced dysfunction (Cochrane)	Meta-analysis	Benefit at 150 mg twice daily	Add or switch to bupropion

★ Putting it together: female pelvic decisions

A · INCONTINENT WOMAN

ALGORITHM

1. **Classify** (history, diary): stress (SUI), urgency (UUI), mixed, overflow, functional. Check stress test, prolapse (POP), residual, urinalysis (Ch 105).
2. **Urodynamics** are not needed for uncomplicated demonstrable SUI (VALUE). Use them if mixed, voiding dysfunction, prior surgery, neurological or high-grade POP.
3. **Conservative first:** weight, fluids, supervised pelvic floor and bladder training, pessary, vaginal oestrogen (Ch 114).
4. If this fails, then **SUI surgery** (Ch 116, 118); for UUI, drugs, botulinum toxin, neuromodulation.

B · CHOOSING AN SUI OPERATION

Operation	Best for	Drawbacks
Retropubic mid-urethral sling (MUS)	Standard; intrinsic sphincter deficiency	Perforation (cystoscopy mandatory), voiding dysfunction
Transobturator MUS	Avoiding the retropubic space	Groin/thigh pain; weaker in severe SUI
Autologous pubovaginal sling	Recurrent SUI, urethral repair, mesh declined	More voiding dysfunction
Burch	Mesh declined; concurrent abdominal surgery	Posterior prolapse, de novo urgency
Bulking agent	Frail, minimal surgery	Lower, short-lived cure

C · PROLAPSE: WHICH REPAIR?

ALGORITHM

1. **POP-Q** stage II is a leading edge within 1 cm of the hymen (Ch 105).
2. Observe if asymptomatic; offer pessary or surgery if symptomatic.
3. **Support the apex**: uterosacral/sacrospinous suspension or sacrocolpopexy (most durable) (Ch 117).
4. If **frail, not sexually active**, then colpocleisis.
5. If **occult SUI**, then discuss a concomitant sling (OPUS: fewer leaks, more complications).

D · VESICOVAGINAL FISTULA

KEY POINT

The first repair has the best chance. Do a double-dye test, upper tract imaging and, if irradiated, biopsy. Post-hysterectomy fistulae are supratrigonal and mostly repaired vaginally; go abdominal for ureteric involvement, large or high fistulae, or failed vaginal repair. Use tension-free, non-overlapping layers, a vascularised flap (Martius or omentum) for poor tissue, then a cystogram (Ch 121).

E · MESH COMPLICATIONS

ALGORITHM

1. If suspected, examine and cystoscope.
2. If exposure is small, then oestrogen with or without trimming; otherwise excise and close.
3. If mesh is in bladder or urethra, then remove and repair without new mesh.
4. Remove only enough to solve the problem, never asymptomatic mesh (Ch 119).